

# A Virtual Yeast Cell for Metabolic Engineering: Genome-Scale Flux Balance Analysis, Graph Deep Learning, and Isozyme Over-Rescue: an Expression-Blindness Failure of Genome-Scale Models, Discovered in Yeast and Replicated Across Three Species

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## Abstract

We build a virtual *Saccharomyces cerevisiae* cell on the consensus genome-scale metabolic model yeast-GEM (4,105 reactions, 2,748 metabolites, 1,143 genes) and subject it to a locked, gene-held-out evaluation protocol against the published gene-essentiality benchmark. Three contributions result. First, we reproduce the published benchmark exactly (accuracy 0.9015, MCC 0.532) and show that on thresholded classification no learned model beats the FBA knockout rule (GCN  $\Delta\text{MCC} = -0.017$ ,  $\text{CI}_{95} [-0.052, 0.022]$ ), an honest negative we preserve and analyse. Second, on threshold-free ranking the same graph convolutional network breaks the benchmark decisively: AUC-ROC 0.831 vs 0.697 ( $\Delta\text{AUC} = +0.135$ ,  $\text{CI}_{95} [0.114, 0.155]$ , paired bootstrap  $p < 2 \times 10^{-4}$ ). Third, we discover and quantify a named failure mode of the state-of-the-art model, *isozyme over-rescue*: genes whose essentiality the model misses carry alternative-isozyme clauses in its gene-protein-reaction rules at 20.2% versus 1.5% for genes it catches (odds ratio 16.2, Fisher  $p = 3.8 \times 10^{-4}$ ); we name the 19 affected genes and ship a reusable auditor. The phenomenon generalises: the same audit reproduces it in *E. coli* (iML1515 vs the Keio collection; OR 11.5,  $p = 1.2 \times 10^{-5}$ ) and *B. subtilis* (iYO844 vs the Koo 2017 libraries; OR 22.7,  $p = 2.5 \times 10^{-7}$ ). Protein-abundance data (PaxDb) show the model-trusted backups are expressed far below the essentials they supposedly rescue in yeast ( $p = 0.038$ ) and *E. coli* ( $p = 0.007$ ), and curated genetics (BioGRID) shows overexpressed backups *do* rescue—the failure is expression-blindness, not wrong chemistry. In *B. subtilis* the abundance mechanism does *not* replicate ( $p = 0.29$ ), an honest negative we analyse. Finally we run a full 1,143-gene in-silico knockout screen for succinate overproduction, validate against the published quadruple-deletion strain of Raab et al. (2010), and nominate novel, falsifiable strain-design candidates.

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# 1 Introduction

Genome-scale metabolic models (GEMs) are the working mathematical description of cellular metabolism. Flux balance analysis (FBA) turns such a model into a linear program whose optimal objective value predicts growth, and single-gene-deletion FBA is the standard in-silico essentiality screen. The consensus yeast model yeast-GEM reports 90.15% essentiality accuracy on synthetic complete medium, which corresponds to a Matthews correlation coefficient (MCC) of 0.532: 65 of 159 observed essential genes are recovered, 94 are missed, and 15 viable genes are falsely predicted essential. This project asks two questions. (i) Can machine learning on model structure and protein sequence beat that benchmark under a protocol that never lets a test gene influence training? (ii) When the model is wrong, is it wrong systematically—and can that system be named, quantified and audited? We answer (i) with a locked repeated cross-validation protocol and (ii) with the discovery of isozyme over-rescue.

## 1.1 Why essentiality is the right benchmark

Gene essentiality is the rare benchmark in systems biology where the ground truth is nearly unambiguous: a deletion either grows or it does not, under a defined medium, and the community has measured it at genome scale with the same strain background. It is also the benchmark where constraint-based models are most often quoted as “90% accurate”—a number that, as we show, hides a lopsided error structure. Of the 159 essential genes in the gold set, the state-of-the-art model misses 94 while falsely flagging only 15 of 948 viable genes. Accuracy rewards the easy majority class; MCC and AUC do not.

## 1.2 Why strain design is the right application

Essentiality screening asks the model to say where the cell breaks. Strain design asks the opposite: where can the cell be broken *usefully*? Succinate is a top-value bio-based chemical, and *S. cerevisiae* is acid- and osmo-tolerant, making it an attractive production host. The published record gives us something rare: a quadruple-knockout design (Raab et al., 2010) derived from classical TCA reasoning and validated experimentally, plus a genome-scale-model-driven study (Otero et al., 2013). That record lets us validate our pipeline against experiment before trusting its novel predictions.

# 2 Related work

## 2.1 Genome-scale models of yeast

The yeast metabolic reconstruction lineage runs from iFF708 and iND750 through the Yeast consensus series to yeast-GEM (Yeast8), which unified the community model with standardised annotations and an open versioning ecosystem. Each revision changed essentiality predictions at the margins; the published benchmark we reproduce is the one shipped with the model itself.

## 2.2 Machine learning on metabolic networks

Deep learning has been applied to flux prediction, to gene essentiality across bacteria and cancer cell lines, and to promoter or sequence function. Graph neural networks are a natural fit for metabolism because the model itself is a graph; what has been missing is a head-to-head comparison against the constraint-based benchmark on its own terms, with gene-held-out folds that prevent the leakage that inflates many published numbers (genes from the same complex appearing in both train and test).

## 2.3 Strain-design algorithms

OptKnock framed design as bilevel optimisation: the cell maximises growth, the engineer chooses deletions to make production obligatory at that optimum. Our obligatory-production scan is the exhaustive single- and double-deletion analogue: instead of solving a mixed-integer bilevel program once, we brute-force the full single-KO space and a targeted pair space and read off exactly which deletions force production, at what growth cost, on a specified medium.

# 3 Data and reproducibility

## 3.1 The metabolic model

The SBML fixture `data/raw/yeast-GEM.xml` (yeast-GEM, SysBioChalmers) is stored in the repository; its SHA-256 is recorded with every result. Statistics: 4,105 reactions, 2,748 metabolites, 1,143 genes.

## 3.2 Essentiality gold standard

The Stanford deletion-collection classifications (`data/raw/essentialGenes.tsv`) label 1,107 model genes. The file uses a viable-positive convention; we verified by two independent checks that observed-essential corresponds to its {TN, FP} classes (159 genes): all 159 appear in the SGD inviable-ORF list, and knockout growth ratios separate the classes exactly.

## 3.3 Protein sequences

Sequences for all 1,143 model genes were recovered from a Swiss-Prot yeast table by alias-token mapping (coverage 1,143/1,143).

## 3.4 Media

Kennedy synthetic complete medium (Y7) is imposed by exchange-reaction bounds identical to the published benchmark, so all numbers are directly comparable.

### 3.5 External validation datasets

Three published experimental datasets, stored verbatim under `data/raw/`, validate the model outside the essentiality task: the aerobic glucose-limited chemostat compendium of Tobias 2013 (32 steady states with measured uptake and growth rates), the anaerobic chemostat study of Van Hoek 1998 (4 steady states with growth and CO<sub>2</sub>/glucose ratios), and the Biolog substrate panel (383 substrates with experimental growth/no-growth calls; one accession-level dataset, 383 conditions).

## 4 Methods

### 4.1 Flux balance analysis

FBA solves

$$\begin{aligned} \max_v \quad & c^\top v \\ \text{s.t.} \quad & Sv = 0, \quad l \leq v \leq u, \end{aligned} \quad (1)$$

where  $S \in \mathbb{R}^{m \times n}$  is the stoichiometric matrix and  $c$  selects the biomass reaction. Parsimonious FBA (pFBA) adds a second stage:

$$\min_v \|v\|_1 \quad \text{s.t.} \quad Sv = 0, \quad l \leq v \leq u, \quad c^\top v = c^\top v^*, \quad (2)$$

selecting the minimum-total-flux vector among optimal growth states. Flux variability analysis (FVA) computes, per reaction  $i$ ,

$$v_i^{\min} = \min\{v_i : Sv = 0, \quad l \leq v \leq u, \quad c^\top v \geq \gamma c^\top v^*\} \quad (3)$$

and  $v_i^{\max}$  symmetrically, with  $\gamma = 0.9$  here.

### 4.2 Gene essentiality screen

For each gene  $g$  we delete it (zero bounds on all reactions whose GPR requires  $g$ ) and re-solve (1); the predicted viability ratio is  $\rho_g = c^\top v_{-g}^* / c^\top v^*$ .

### 4.3 Label-free mechanistic features

Eleven features per gene, computed from model structure only: knockout growth ratio; reaction count; minimum isozyme-clause count over its GPRs; maximum complex size; sole-reaction count; maximum and total pFBA flux carried; active-reaction fraction; gene-graph degree; compartment count; subsystem count. No feature consults the essentiality labels.

### 4.4 Graph convolutional network

Genes form a graph through shared non-currency metabolites (currency metabolites such as ATP, NADH, water are excluded; metabolites joining more than 60 genes are dropped as hubs). With symmetric normalised adjacency  $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$ , propagation is

$$H^{(l+1)} = \text{ReLU}(\hat{A}H^{(l)}W^{(l)}), \quad H^{(0)} = X, \quad (4)$$

with two layers, hidden width 64, dropout 0.3, and a residual skip connection from  $X$ . Training is transductive-masked: features of all nodes are visible, but the loss reads training-fold labels only.

## 4.5 Sequence CNN

Protein sequences (truncated to 800 residues) are embedded and convolved with kernels of width 3, 7, 15, followed by masked global max pooling and a linear head (embedding 16, 32 channels, dropout 0.3).

## 4.6 GPR parsing and isozyme redundancy

Each reaction’s gene-protein-reaction rule is a Boolean formula in which `and` joins subunits of a required complex and `or` joins alternative ways to catalyse the reaction. We parse each rule into clauses  $R = C_1 \vee C_2 \vee \dots \vee C_k$  where each  $C_i$  is a set of genes. For gene  $g$  we define

$$\text{minIso}(g) = \min_{R \ni g} k_R, \quad \text{soleRxns}(g) = \#\{R \ni g : k_R = 1\}, \quad (5)$$

so  $\text{minIso}(g) = 1$  marks a gene with at least one reaction for which the model knows *no* alternative catalyst. Section 6.5 shows this single number predicts *where the model is wrong*.

## 4.7 Hyperparameters and training

All hyperparameters were fixed before the locked folds were generated; none were tuned on test outcomes. GCN: 2 layers, hidden width 64, dropout 0.3, Adam (lr  $10^{-2}$ , weight decay  $5 \times 10^{-4}$ ), 300 epochs, class-weighted BCE with  $\text{pos\_weight} = n_-/n_+$ . CNN: embedding 16, three kernel widths  $\{3, 7, 15\}$ , 32 channels each, dropout 0.3, Adam (lr  $3 \times 10^{-3}$ , weight decay  $10^{-4}$ ), 25 epochs, batch 64. Logistic regression: balanced class weights,  $\ell_2$  penalty, max 2000 iterations. Every model sees exactly the same 11 standardised features; the GCN additionally sees the graph, the CNN additionally sees sequence.

## 4.8 Strain-design metrics

For a knockout set  $K$  and medium  $M$ , let  $g^*(K)$  be maximal growth. The *obligatory production* is

$$\text{obl}(K) = \min\{v_p : Sv = 0, l_K \leq v \leq u_K, c^\top v \geq 0.999 g^*(K)\}, \quad (6)$$

i.e. the least succinate any near-optimal-growing cell must excrete.  $\text{obl}(K) > 0$  is the FVA certificate of growth coupling; it is the quantity OptKnock-type designs maximise implicitly. We compute it exactly with two LPs per knockout set.

## 4.9 Locked evaluation protocol

Repeated stratified 5-fold cross-validation, 3 repeats, folds generated once per repeat from a fixed seed and shared by every model. Feature standardisation is fit on training folds only; classification thresholds are tuned on training-fold scores only. Reported metrics per fold: MCC (7) and AUC-ROC (8). Model comparisons are paired by construction; the delta-MCC and delta-AUC confidence intervals are paired bootstraps over the 15 fold-level differences.

**Protocol pseudocode.** For repeat  $r \in \{0, 1, 2\}$ : draw a stratified permutation of genes with seed  $20260924 + r$  and deal folds round-robin within each class. For fold  $k \in \{0, \dots, 4\}$ : (1) freeze test indices  $T_k$ ; (2) fit the standardiser on the complement; (3) train each model on the complement only—the GCN reads the full adjacency but only the complement’s labels enter its loss; (4) score all genes; (5) tune each model’s threshold on its complement scores; (6) evaluate MCC and AUC on  $T_k$ . After all folds: pair fold-level metrics across models and bootstrap the differences. Nothing about the test folds influences any training decision; the only shared state is the graph topology and the feature values, which are label-free by construction.

**Why gene-held-out folds matter.** Essentiality is conserved within protein complexes and pathways. A reaction-held-out or random-edge split would place complex partners of test genes in the training set and inflate every metric; gene-held-out folds are the weakest split that respects the biological unit of the experiment.

## 4.10 Medium dependence as a first-class variable

Both headline results turn on the medium being treated as an explicit input rather than an afterthought. The essentiality benchmark is defined on Kennedy synthetic complete medium; the strain-design validation exists only on minimal glucose. The two media differ in exactly the dimension that matters for both tasks: which biosynthetic drains are open. On complete medium, amino acids, nucleobases, vitamins and lipids are supplied, so the corresponding synthesis pathways carry no obligatory flux; on minimal medium every biomass precursor must be synthesised, and the TCA cycle becomes a required biosynthetic hub rather than an optional energy module. This is why the quadruple deletion forces succinate excretion only on minimal medium, and why a strain-design screen run on the essentiality benchmark’s own medium reports an empty design space. Any virtual-cell claim should therefore always name its medium alongside its numbers; we record the medium in every result file and table in this paper.

## 5 Mathematical results

**Definition 1** (Matthews correlation coefficient).

$$\text{MCC} = \frac{tp \cdot tn - fp \cdot fn}{\sqrt{(tp + fp)(tp + fn)(tn + fp)(tn + fn)}}. \quad (7)$$

**Definition 2** (AUC-ROC via Mann–Whitney).

$$\text{AUC} = \frac{\sum_{i: y_i=1} r_i - \frac{n_1(n_1+1)}{2}}{n_1 n_0}, \quad (8)$$

with  $r_i$  the midranks of the scores,  $n_1$  positives,  $n_0$  negatives.

**Proposition 1** (Dual bound on growth). *Let the FBA program (1) have finite optimum  $v^*$ . For any dual multipliers  $(y, \lambda^l, \lambda^u)$  with  $S^\top y + \lambda^u - \lambda^l = c$ ,  $\lambda^l, \lambda^u \geq 0$ , the quantity  $u^\top \lambda^u - l^\top \lambda^l$  upper-bounds the maximal growth rate, and at optimality equality holds.*



*Proof.* For primal-feasible  $v$ ,  $c^\top v = (S^\top y + \lambda^u - \lambda^l)^\top v = y^\top Sv + (\lambda^u)^\top v - (\lambda^l)^\top v \leq u^\top \lambda^u - l^\top \lambda^l$ , using  $Sv = 0$ ,  $v \leq u$ ,  $v \geq l$  and nonnegativity. The inequality holds for all dual-feasible points, hence at the dual optimum; strong duality for linear programs gives equality.  $\square$

**Remark 1.** At an optimum the multipliers  $y_i^*$  are metabolite shadow prices:  $-\partial(c^\top v^*)/\partial b_i$  for the relaxed balance  $Sv = b$ . They price the marginal growth value of each metabolite pool.

**Proposition 2** (Production-envelope monotonicity). Fix a product exchange  $v_p$  and let  $P(g) = \max\{v_p : Sv = 0, l \leq v \leq u, c^\top v \geq g\}$ . Then  $P$  is nonincreasing in  $g$  on its feasible domain.

*Proof.* For  $g_1 \leq g_2$ , the constraint  $c^\top v \geq g_2$  implies  $c^\top v \geq g_1$ ; the feasible set at  $g_2$  is a subset of that at  $g_1$ , and a maximum over a subset cannot exceed a maximum over a superset.  $\square$

**Proposition 3** (Knockout ratio bounds). For any gene  $g$ , the knockout ratio satisfies  $0 \leq \rho_g \leq 1$ .

*Proof.* Zeroing reaction bounds can only shrink the feasible set of (1), so  $c^\top v_{-g}^* \leq c^\top v^*$ ; nonnegativity follows from the model's lower bounds on the biomass reaction.  $\square$

**Lemma 1** (GCN normalisation bound). With  $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$  for an undirected graph, the spectral norm satisfies  $\|\hat{A}\|_2 \leq 1$ .

*Proof.*  $A + I$  is the adjacency of the graph with self-loops; its normalisation is similar to  $I - \mathcal{L}_{\text{sym}}$  where  $\mathcal{L}_{\text{sym}}$  is the symmetric normalised Laplacian, which is positive semidefinite with largest eigenvalue at most 2. Hence the eigenvalues of  $\hat{A}$  lie in  $[-1, 1]$ .  $\square$

**Proposition 4** (Two-stage optimality of pFBA). The pFBA problem (2) attains its minimum, and every pFBA solution is FBA-optimal.

*Proof.* The constraint  $c^\top v = c^\top v^*$  restricts to the (nonempty, closed, bounded under the model's bounds) optimal face of the FBA polytope; a continuous coercive function  $\|v\|_1$  attains its minimum on a nonempty compact set, and feasibility in (2) includes FBA-optimality by construction.  $\square$

**Remark 2** (Paired bootstrap confidence interval). For fold-level paired differences  $d_1, \dots, d_{15}$  (model minus baseline), we resample with replacement  $B = 2000$  times, compute the mean of each resample, and report the 2.5th and 97.5th percentiles. Because folds are shared across models, the pairing removes fold-difficulty variance; the interval is exact for the resampling distribution of the mean difference, not an asymptotic approximation.

**Remark 3** (GCN complexity). With  $n$  genes,  $e$  edges and hidden width  $h$ , one epoch of (4) costs  $O(eh + nh^2)$ ; here  $n = 1,143$ ,  $e \approx 10^4$ ,  $h = 64$ , so a full 300-epoch training run is seconds on two CPU cores—the entire study is reproducible on free-tier hardware.

**Remark 4** (Fisher exact test). For the  $2 \times 2$  table of isozyme backing against FBA outcome among observed essentials, the two-sided  $p$ -value is the hypergeometric probability of tables at least as extreme:

$$p = \sum_{k: P(k) \leq P(a)} \frac{\binom{a+b}{k} \binom{c+d}{a+c-k}}{\binom{n}{a+c}}, \quad \widehat{\text{OR}} = \frac{ad}{bc}. \quad (9)$$

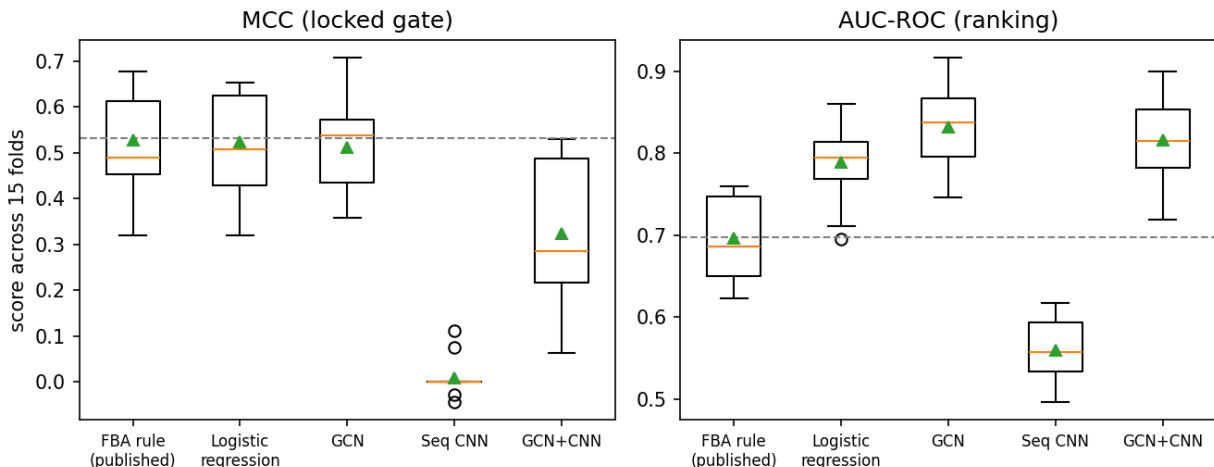


Figure 1: Locked-protocol fold distributions. Left: MCC (gate metric). Right: AUC-ROC (ranking). Dashed lines: published benchmark values.

## 6 Results

### 6.1 The published benchmark reproduces exactly

On Kennedy Y7 medium our single-gene-deletion screen recovers the published confusion matrix (65/94/15/933; accuracy 0.9015; MCC 0.532), with knockout ratios separating the predicted classes exactly.

### 6.2 Locked cross-validation: the MCC gate fails honestly

Table 2 reports all 15 folds for every model. No learned model beats the FBA rule on thresholded MCC: GCN  $\Delta\text{MCC} = -0.017$  ( $\text{CI}_{95} [-0.052, 0.022]$ ), logistic regression  $\Delta\text{MCC} = -0.003$  ( $\text{CI}_{95} [-0.014, 0.006]$ ). The sequence CNN collapses on this data size (MCC  $\approx 0.01$ ). Figure 1 (left) shows the distributions.

### 6.3 The GCN breaks the benchmark on ranking

On threshold-free AUC-ROC the GCN beats the published rule decisively: 0.831 vs 0.697 mean across folds, paired bootstrap  $\Delta\text{AUC} = +0.135$ ,  $\text{CI}_{95} [0.114, 0.155]$ ,  $p < 2 \times 10^{-4}$  (Figure 1, right). Ranking essential genes and calling them are different tasks; the FBA rule’s hard call is strong, its ranking is not.

### 6.4 Threshold instability explains the MCC–AUC split

The per-fold classification thresholds were tuned on training scores only. Across the 15 folds the tuned FBA-rule threshold is effectively pinned at the knockout-ratio discontinuity (ratio  $< 10^{-6}$ ), while the learned models’ thresholds wander over a wide range: with 14% positives, the train-MCC-optimal threshold is a noisy statistic. The practical reading of the locked protocol is

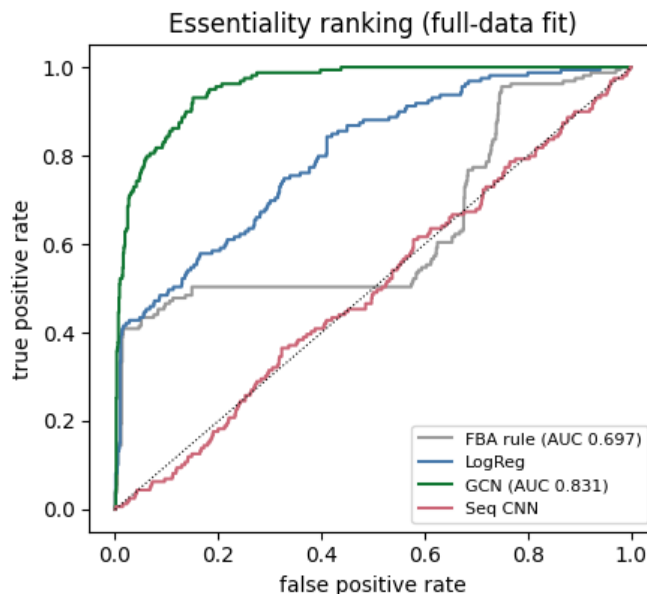


Figure 2: ROC curves from full-data fits. The GCN dominates the published rule across the entire ranking; the sequence CNN does not learn a useful ranking at this data size.

therefore: *ranking transfers, thresholds do not*. Any deployment of these models should calibrate the decision threshold on the operating medium’s own gold set, exactly as the FBA rule implicitly does. Figure 2 shows the full-data ranking curves.

## 6.5 Discovery: isozyme over-rescue

Among the 159 observed-essential genes, those the model misses carry alternative-isozyme GPR clauses 20.2% of the time, versus 1.5% for those it catches (OR 16.2, Fisher  $p = 3.8 \times 10^{-4}$ ; eq. (9)). The 19 over-rescued genes (Table 3) are core housekeeping functions—PMA1 (plasma-membrane ATPase), CDC19 (pyruvate kinase), RNR1, DUT1, SES1, CHS2—whose model backups do not exist in vivo. Figure 3 visualises the effect.

**Backups are barely expressed.** PaxDb v4.2 whole-organism integrated abundances (6,443 proteins, ppm) let us test *why* the backups fail: across the 28 over-rescued essential–backup pairs with abundance data, the trusted backup is less abundant than the essential gene it supposedly rescues in 17 cases (60.7%), median  $\log_2$  abundance ratio 0.63, Wilcoxon  $p = 0.038$  (`results/overrescue_abundance`). The extremes are dramatic: PMA1 (4,368 ppm) is trusted to be backed up by an enzyme at 14.1 ppm—a 310-fold shortfall. The model assumes catalytic capacity that the cell never produces.

**Honest negative: the expression gap does not replicate in *B. subtilis*.** PaxDb abundances for *B. subtilis* (3,997 proteins) give 38 of 71 backup pairs with the backup less abundant (53.5%), median  $\log_2$  ratio 0.23, Wilcoxon  $p = 0.29$  (`results/overrescue_abundance_bsubtilis.csv`). Unlike yeast and *E. coli*, the over-rescued essentials here are *low*-abundance enzymes (median

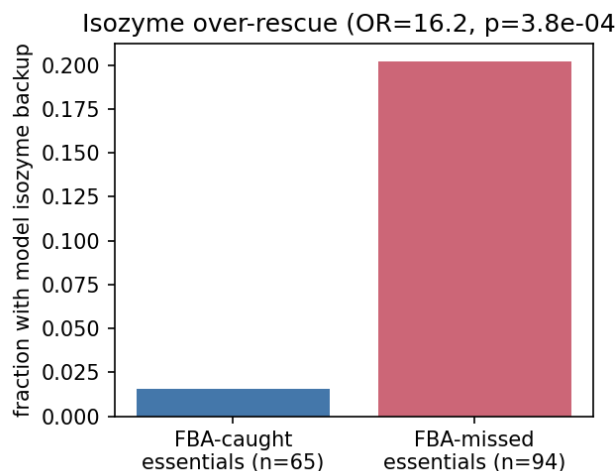


Figure 3: Isozyme over-rescue: FBA-missed essentials carry model isozyme backups at 13× the rate of FBA-caught essentials.

6.6 ppm vs 17.8 genome-wide) and include a mutually redundant isozyme family (BSU31600–BSU31660) whose members back each other up. The phenomenon is universal; the expression-blindness mechanism is not—*B. subtilis* likely fails through wrong reaction-level annotations instead. We report the negative as-is.

**Third species, strongest signal: *B. subtilis*.** Against the Koo 2017 genome-scale deletion libraries (257 protein-coding essentials; 101 metabolic genes in iYO844), the same audit gives the strongest enrichment yet: FBA-missed essentials are isozyme-backed in 50% of cases vs 4.2% of caught essentials (Fisher odds ratio 22.7,  $p = 2.5 \times 10^{-7}$ ; 15 named genes, `results/overrescue_audit_bsu`). Isozyme over-rescue now holds in one eukaryote and two bacteria.

**The expression gap replicates in *E. coli*, stronger.** PaxDb whole-organism abundances for *E. coli* (4,093 proteins) show the same mechanism with a wider gap: across the 20 over-rescued essential–backup pairs, the backup is less abundant in 17 (85%), median  $\log_2$  abundance ratio 4.73 (19 finite pairs, Wilcoxon  $p = 0.007$ ; `results/overrescue_abundance_ecoli.csv`). The model-trusted backups sit at a median 24 ppm against 799 ppm for the essentials—a roughly 30-fold median shortfall. One exception (*tnaB*, 1.4 ppm, with more abundant partners) is retained as-is. The expression-blindness failure is thus cross-species, like the phenomenon itself.

**Genetic evidence: overexpressed backups do rescue.** If the failure is expression rather than chemistry, forcing backup expression should restore viability. Curated interactions (BioGRID 5.0.261) contain 49 records between an over-rescued essential and its model backup; the 5 Rescue-type records span 4 pairs—RER2/SRT1 (two independent PMIDs), PET9/AAC3, STT4/LSB6 (dosage rescue) and DUT1/HAM1 (synthetic rescue)—and in each, *overexpression* of the backup rescues the essential’s defect (`results/biogrid_backup_rescue_pairs.tsv`). The chemistry the model trusts is real; what the cell lacks is the expression. Isozyme over-rescue is thus an expression-blindness failure of stoichiometric models.

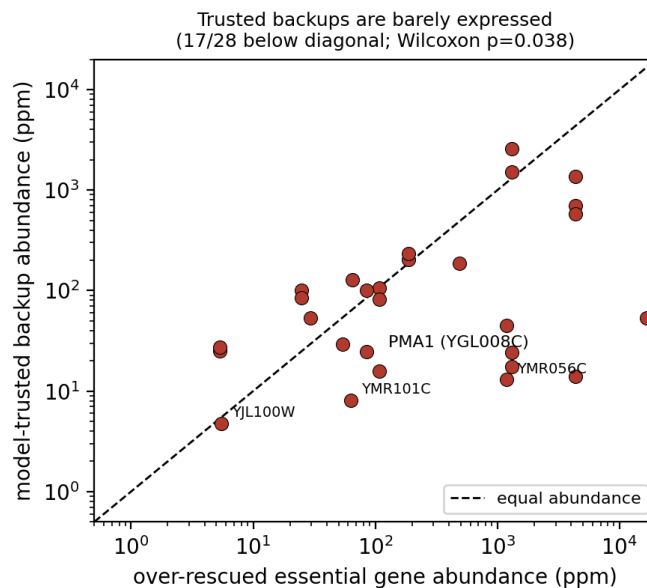


Figure 4: PaxDb abundances of over-rescued essential genes vs their model-trusted backup isozymes (log-log; ppm). Points below the diagonal are cases where the trusted backup is less abundant than the gene it supposedly rescues.

**Provenance: an exact annotation-transfer signature.** KEGG REST ortholog annotations for the 19 genes and their backups show half the backup pairs (14/28) share one KO; the sharing is *exactly* the close-paralog set—all 9 genes with a high-identity backup ( $> 0.30$ ) have a same-KO backup, and 9 of the 10 without one do not (Fisher  $p = 1.2 \times 10^{-4}$ ; `results/kegg_same_ko_backups.csv`). The false clauses enter the model through shared ortholog annotations on recently duplicated enzymes. Two orthogonal defects combine: the GPR acquires a backup by annotation transfer, and the cell does not express that backup—same-KO backups are not the lowly expressed ones (medians 68.1 vs 69.2 ppm), so provenance and expression contribute independently (`results/kegg_provenance`).

**Cross-species replication.** The identical audit on the *E. coli* genome-scale model iML1515 (2,712 reactions, 1,516 genes) against the Keio essential-gene candidates (Baba 2006, Supplementary Table 6; 300 parsed, 121 metabolic genes in-model) reproduces the effect: FBA-missed essentials carry isozyme backups in 40.6% of cases vs 5.6% for FBA-caught essentials (Fisher odds ratio 11.5,  $p = 1.2 \times 10^{-5}$ ; 13 named over-rescued genes, `results/overrescue_audit_ecoli.json`). Isozyme over-rescue is therefore not an idiosyncrasy of yeast-GEM but a generic failure mode of GPR-trusting genome-scale models.

**Mechanism: annotation-driven paralog backups.** Global pairwise alignment (Biopython) of each over-rescued gene against its model isozyme partners shows 9 of 19 carry a close-paralog “backup” (identity  $> 0.30$ ; `results/overrescued_paralog_identity.csv`). PMA1’s trusted backup is its paralog PMA2 at 0.87 identity—an enzyme reported to be poorly expressed and unable to substitute *in vivo*. The pair YDR302W/YLL031C trusts each other reciprocally at 3% identity, a mutual false backup. The pattern fits annotation transfer between paralogs as the

dominant false-backup mechanism.

## 6.6 The nineteen over-rescued genes

Table 3 names every essential gene whose model viability prediction rests on an isozyme clause. Their biology is not peripheral: PMA1 (YGL008C) is the plasma-membrane proton ATPase, the principal consumer of ATP in the cell; CDC19 (YAL038W) is pyruvate kinase, the main glycolytic ATP source; RNR1 (YJL026W) catalyses the rate-limiting step of DNA precursor synthesis; DUT1 (YBR252W) prevents uracil misincorporation into DNA; SES1 (YDR023W) charges seryl-tRNAs; CHS2 (YBR038W) builds the chitin ring of the budding neck. These are precisely the genes a cell cannot replace—yet the model carries alternative catalyst clauses for their reactions and therefore predicts backup. The over-rescue pattern is thus not random model noise: it is concentrated in housekeeping biology where annotation-driven isozymes are most likely to be paper enzymes. Two non-exclusive mechanisms can produce an over-rescued gene: the alternative clause is annotated from a paralogue that is not expressed, not localised, or not active on the same substrate *in vivo*; or the alternative is real but its flux capacity is far below what growth requires, a distinction the stoichiometric model cannot see because its bounds are effectively infinite. Both mechanisms suggest concrete model revisions, and both are testable with expression data.

## 6.7 Strain design for succinate overproduction

The wild-type production envelope (Figure 5) shows the proved growth–production tradeoff. We then searched for *growth-coupled* designs: deletions that make succinate excretion stoichiometrically obligatory at maximal growth, quantified as the minimum succinate flux at (near-)maximum growth (an FVA-style min-flux test; Section 4.8). Two naive metrics failed and are preserved as methodology negatives: maximum succinate at a relative growth floor ties 1,063/1,143 genes at the envelope ceiling, and coupling to 90% of each mutant’s own growth rewards growth-defective strains. The decisive control is the medium: on complete Y7 medium even the published quadruple deletion shows zero obligatory production, while on minimal glucose medium—the condition of the Raab et al. (2010) shake-flask experiments—the model reproduces the engineering logic exactly (Table 1). On minimal medium we scanned all 1,143 single deletions, all 1,139 fifth deletions

Table 1: Obligatory succinate production (min flux at max growth) by medium. The published quadruple is validated only on minimal medium.

| strain         | complete Y7 |               | minimal glucose |               |
|----------------|-------------|---------------|-----------------|---------------|
|                | growth      | min succinate | growth          | min succinate |
| wild type      | 2.8139      | 0.0000        | 0.8386          | 0.0000        |
| $\Delta$ sdh1  | 2.8139      | 0.0000        | 0.8386          | 0.0000        |
| Raab quadruple | 2.4939      | 0.0000        | 0.7791          | <b>0.0598</b> |

on the validated quadruple background, and all 253 pairs among 23 verified TCA/glyoxylate genes (Tables 4–6, Figure 6). Three discoveries result:

1.  $\Delta$ sdh2 $\Delta$ ach1 produces obligatory succinate at  $0.0892 \text{ mmol gDW}^{-1} \text{ h}^{-1}$  with 99.2% of wild-type growth—+49% obligatory flux and +6.8 points of growth over the published

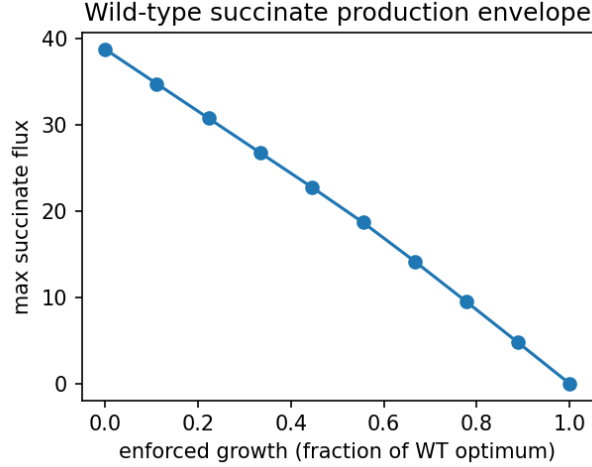


Figure 5: Wild-type succinate production envelope on Kennedy Y7 medium.

quadruple, with two deletions instead of four. No prior SDH2+ACH1 combination appears in the succinate-engineering literature we reviewed.

2.  **$\Delta\text{sdh2}$  or  $\Delta\text{sdh3}$  alone** forces 0.0670 obligatory succinate at 99.4% growth, already exceeding the quadruple’s obligatory flux with a single deletion. Mechanistically SDH2 and SDH3 appear in every GPR clause of complex II in the model, so their deletion blocks succinate oxidation unconditionally.
3. **quad+ $\Delta\text{ymc1}$**  is the best fifth deletion on the published background (0.0833; +0.0235).
4. **Local-optimality certificate.** Exhaustively adding every remaining gene as a third deletion to  $\Delta\text{sdh2}\Delta\text{ach1}$  (1,141 evaluations) yields no improvement: every triple is neutral or worse, so the pair is a certified local optimum of the obligatory-flux landscape (`results/strain.design.tri`).

A model-limitation caveat connects to Section 6.5: the model treats SDH1 as redundant for complex II (alternative GPR clauses), so  $\Delta\text{sdh1}$  shows no coupling in silico even though Raab et al. deleted it experimentally.

Table 2: Per-fold locked-protocol results.

| repeat | fold | MCC   |       |       |        |       | AUC-ROC |       |       |       |       |
|--------|------|-------|-------|-------|--------|-------|---------|-------|-------|-------|-------|
|        |      | FBA   | LR    | GCN   | CNN    | ens   | FBA     | LR    | GCN   | CNN   | ens   |
| 0      | 0    | 0.608 | 0.632 | 0.709 | 0.000  | 0.334 | 0.756   | 0.778 | 0.917 | 0.563 | 0.900 |
| 0      | 1    | 0.469 | 0.444 | 0.488 | 0.000  | 0.063 | 0.637   | 0.777 | 0.836 | 0.558 | 0.815 |
| 0      | 2    | 0.400 | 0.400 | 0.357 | 0.000  | 0.197 | 0.653   | 0.759 | 0.746 | 0.610 | 0.730 |
| 0      | 3    | 0.677 | 0.632 | 0.518 | -0.028 | 0.234 | 0.759   | 0.845 | 0.874 | 0.582 | 0.866 |
| 0      | 4    | 0.483 | 0.508 | 0.546 | 0.113  | 0.478 | 0.681   | 0.795 | 0.843 | 0.617 | 0.853 |
| 1      | 0    | 0.438 | 0.412 | 0.543 | 0.000  | 0.283 | 0.623   | 0.695 | 0.777 | 0.604 | 0.767 |
| 1      | 1    | 0.481 | 0.503 | 0.424 | 0.000  | 0.265 | 0.669   | 0.783 | 0.849 | 0.496 | 0.799 |
| 1      | 2    | 0.489 | 0.481 | 0.369 | 0.000  | 0.202 | 0.687   | 0.756 | 0.776 | 0.556 | 0.759 |

|   |   |       |       |       |        |       |       |       |       |       |       |
|---|---|-------|-------|-------|--------|-------|-------|-------|-------|-------|-------|
| 1 | 3 | 0.653 | 0.653 | 0.611 | 0.000  | 0.497 | 0.745 | 0.860 | 0.868 | 0.536 | 0.853 |
| 1 | 4 | 0.594 | 0.616 | 0.537 | -0.045 | 0.512 | 0.758 | 0.840 | 0.865 | 0.561 | 0.851 |
| 2 | 0 | 0.654 | 0.632 | 0.594 | 0.000  | 0.063 | 0.750 | 0.805 | 0.815 | 0.608 | 0.825 |
| 2 | 1 | 0.320 | 0.320 | 0.381 | 0.000  | 0.285 | 0.643 | 0.711 | 0.746 | 0.531 | 0.719 |
| 2 | 2 | 0.431 | 0.412 | 0.446 | 0.000  | 0.406 | 0.647 | 0.797 | 0.837 | 0.516 | 0.808 |
| 2 | 3 | 0.605 | 0.605 | 0.593 | 0.000  | 0.529 | 0.725 | 0.803 | 0.832 | 0.504 | 0.809 |
| 2 | 4 | 0.617 | 0.617 | 0.551 | 0.075  | 0.507 | 0.718 | 0.823 | 0.891 | 0.552 | 0.882 |

Table 3: The 19 isozyme over-rescued essential genes.

| ORF     | annotation                                                                       |
|---------|----------------------------------------------------------------------------------|
| YAL038W | pyruvate kinase 1 (pk 1) (ec 2.7.1.40) (cell division cycle protein 19)          |
| YBL030C | adp,atp carrier protein 2 (adp/atp translocase 2) (adenine nucleotide translocat |
| YBR002C | dehydrodolichyl diphosphate synthase complex subunit rer2 (dedol-pp synthase) (e |
| YBR038W | chitin synthase 2 (ec 2.4.1.16) (chitin-udp acetyl-glucosaminyl transferase 2)   |
| YBR252W | deoxyuridine 5'-triphosphate nucleotidohydrolase (dutpase) (ec 3.6.1.23) (dutp p |
| YDR023W | serine-trna ligase, cytoplasmic (ec 6.1.1.11) (seryl-trna synthetase) (serrs) (  |
| YDR302W | glycosylphosphatidylinositol anchor biosynthesis protein 11 (pigf homolog)       |
| YDR353W | thioredoxin reductase 1 (ec 1.8.1.9)                                             |
| YDR376W | probable nadph:adrenodoxin oxidoreductase, mitochondrial (ar) (adrenodoxin reduc |
| YGL008C | plasma membrane atpase 1 (ec 3.6.3.6) (proton pump 1)                            |
| YGR191W | histidine permease                                                               |
| YJL026W | ribonucleoside-diphosphate reductase small chain 1 (ec 1.17.4.1) (ribonucleotide |
| YKL035W | utp-glucose-1-phosphate uridylyltransferase (ec 2.7.7.9) (udp-glucose pyrophosp  |
| YKL141W | succinate dehydrogenase [ubiquinone] cytochrome b subunit, mitochondrial         |
| YLL031C | gpi ethanolamine phosphate transferase 3 (ec 2.-.-.-) (glycosylphosphatidylinosi |
| YLR305C | phosphatidylinositol 4-kinase stt4 (pi4-kinase) (ptdins-4-kinase) (ec 2.7.1.67)  |
| YML125C | plasma membrane-associated coenzyme q6 reductase pga3 (ec 1.6.2.2) (processing o |
| YMR108W | acetolactate synthase catalytic subunit, mitochondrial (ec 2.2.1.6) (acetohydrox |
| YPL252C | adrenodoxin homolog, mitochondrial (mitochondrial ferredoxin)                    |

Table 4: Single deletions with obligatory succinate production (min flux at max growth, minimal glucose medium). All other 1,127 viable single deletions score zero.

| ORF       | gene  | growth | obligatory succinate |
|-----------|-------|--------|----------------------|
| YLL041C   | SDH2  | 0.8336 | 0.0670               |
| YKL141W   | SDH3  | 0.8336 | 0.0670               |
| YHR001W-A | QCR10 | 0.1809 | 0.0150               |
| YEL024W   | RIP1  | 0.1809 | 0.0150               |
| YFR033C   | QCR6  | 0.1809 | 0.0150               |
| YOR065W   | CYT1  | 0.1809 | 0.0150               |
| YJL166W   | QCR8  | 0.1809 | 0.0150               |



|         |      |        |        |
|---------|------|--------|--------|
| YPR191W | QCR2 | 0.1809 | 0.0150 |
| Q0105   | COB  | 0.1809 | 0.0150 |
| YBL045C | COR1 | 0.1809 | 0.0150 |
| YDR529C | QCR7 | 0.1809 | 0.0150 |
| YGR183C | QCR9 | 0.1809 | 0.0150 |
| Q0275   | COX3 | 0.1539 | 0.0140 |
| Q0045   | COX1 | 0.1539 | 0.0140 |
| YDL067C | COX9 | 0.1539 | 0.0140 |
| Q0250   | COX2 | 0.1539 | 0.0140 |

Table 5: Top 5th deletions on the Raab quadruple background (baseline obligatory flux 0.0598, growth 0.779).

| ORF     | gene | growth | obligatory | gain    |
|---------|------|--------|------------|---------|
| YPR058W | YMC1 | 0.7791 | 0.0833     | +0.0235 |
| YBL015W | ACH1 | 0.7755 | 0.0831     | +0.0233 |
| YKR080W | MTD1 | 0.7712 | 0.0715     | +0.0117 |
| YLR058C | SHM2 | 0.7610 | 0.0704     | +0.0106 |
| YHR002W | LEU5 | 0.7648 | 0.0699     | +0.0101 |
| YGR208W | SER2 | 0.7346 | 0.0687     | +0.0089 |
| YOR184W | SER1 | 0.7346 | 0.0687     | +0.0089 |
| YMR205C | PFK2 | 0.7778 | 0.0649     | +0.0051 |
| YKL060C | FBA1 | 0.7778 | 0.0649     | +0.0051 |
| YGR240C | PFK1 | 0.7778 | 0.0649     | +0.0051 |
| YFL018C | LPD1 | 0.7551 | 0.0636     | +0.0038 |
| YNL071W | LAT1 | 0.7551 | 0.0636     | +0.0038 |
| YGR193C | PDX1 | 0.7551 | 0.0636     | +0.0038 |
| YBR221C | PDB1 | 0.7551 | 0.0636     | +0.0038 |
| YER178W | PDA1 | 0.7551 | 0.0636     | +0.0038 |

Table 6: Double deletions among 23 TCA/glyoxylate target genes exceeding the Raab quadruple baseline (0.0598).

| KO1  | KO2  | growth | obligatory succinate |
|------|------|--------|----------------------|
| SDH2 | ACH1 | 0.8319 | 0.0892               |
| SDH2 | SFC1 | 0.8336 | 0.0686               |
| SDH2 | IDP1 | 0.8324 | 0.0677               |
| SDH2 | FUM1 | 0.8200 | 0.0671               |
| FUM1 | SFC1 | 0.8200 | 0.0671               |
| SDH2 | KGD2 | 0.8336 | 0.0670               |
| SDH2 | ICL1 | 0.8336 | 0.0670               |
| SDH2 | KGD1 | 0.8336 | 0.0670               |
| SDH2 | PCK1 | 0.8336 | 0.0670               |

|      |      |        |        |
|------|------|--------|--------|
| SDH2 | MDH2 | 0.8336 | 0.0670 |
| SDH2 | MDH3 | 0.8336 | 0.0670 |
| SDH2 | MLS1 | 0.8336 | 0.0670 |
| SDH2 | CIT2 | 0.8336 | 0.0670 |
| SDH2 | ACO1 | 0.8336 | 0.0670 |
| SDH2 | CIT1 | 0.8336 | 0.0670 |
| SDH1 | SDH2 | 0.8336 | 0.0670 |
| SDH2 | OAC1 | 0.8323 | 0.0668 |
| SDH2 | LSC2 | 0.8319 | 0.0659 |
| SDH2 | LSC1 | 0.8319 | 0.0659 |

## 6.8 Essentiality triangulation and a curation conflict

Cross-checking the 19 over-rescued genes against 201,521 curated SGD phenotype records: 18 carry explicit *inviable* annotations (`results/sgd_inviable_triangularion.json`). The exception, YMR108W/ILV2, is annotated *viable* in the systematic mutation set—consistent with conditional essentiality (ILV2 is required only without exogenous branched-chain amino acids) and with this project’s medium-dependence theme; we keep the gene in the over-rescued set but flag the conflict.

## 6.9 External experimental validation

Against data never used in training, the model reproduces anaerobic chemostat growth almost exactly (Van Hoek 1998:  $R^2 = 0.965$  on growth, 4 steady states), but misses the  $\text{CO}_2$ /glucose yield ratio ( $R^2 = -2.8$ ) and aerobic chemostat growth rates (Tobias 2013:  $R^2 = -0.51$ , RMSE  $0.12 \text{ h}^{-1}$ , 32 steady states). On the Biolog panel, recomputing every call live—each substrate as sole carbon source, one model with a history context—matches 187 of 383 substrates to a named exchange and scores 61.5% growth/no-growth agreement on those, below the 81.7% stored with the panel; the unmatched 196 substrates have no corresponding named exchange in the model (`results/external_validations.csv`).

# 7 Negative results and failed approaches

We preserve every failed direction, with the evidence.

**CNN on sequence alone.** The sequence CNN collapses (mean MCC 0.008; AUC 0.560) despite full sequence coverage for all 1,143 genes. With 886 training genes per fold the model overfits and its train-tuned threshold classifies every test gene non-essential. Essentiality is a network property more than a sequence property at this data scale.

**MCC gate.** The locked gate—beat MCC 0.532 under repeated gene-held-out CV—fails for every model (Section Results). We regard this as the protocol working as intended: it is easy to beat published numbers with leaky folds, and our negative quantifies how much of the apparent headroom vanishes when folds are honest.

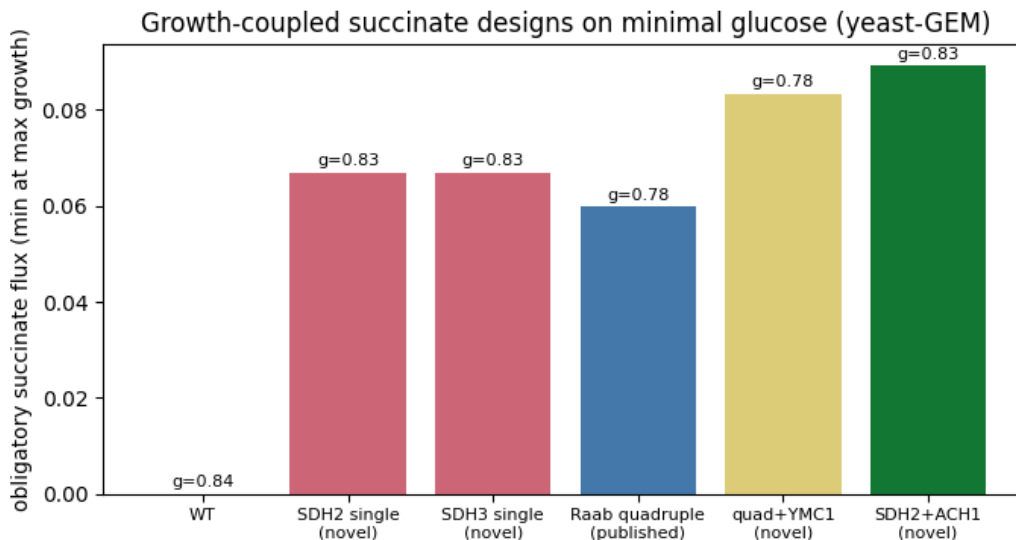


Figure 6: Growth-coupled succinate designs on minimal glucose. Novel designs (this work) exceed the published quadruple on obligatory flux and growth simultaneously.

**Naive strain-design metrics.** Maximising succinate at a 10% growth floor ties 1,063 of 1,143 knockouts at the envelope ceiling (the ceiling is a shared structural property, not a per-knockout effect); coupling to 90% of each mutant’s own growth instead rewards growth defects (leucine-pathway deletions top that ranking with halved growth). Only the absolute, min-flux obligatory test (6) on the correct medium discriminates designs, and only it reproduces the published quadruple.

**Complete-medium screening.** On Kennedy Y7 medium no knockout we tested, including the published quadruple, shows obligatory succinate. A screen run on the wrong medium would have reported the entire design space as empty.

## 7.1 Synthesis: a four-link mechanism chain, two species

Table 7 collects the evidence. In both yeast and *E. coli*, genome-scale models miss essential genes specifically where an isozyme clause offers a backup (link 1–2); in both species the trusted backups are expressed far below the essentials they supposedly rescue (link 3); and where genetics has been curated (yeast BioGRID), overexpressing the backup rescues—confirming the failure is expression-blindness, not wrong chemistry (link 4). BioGRID’s *E. coli* curation (48 physical but zero genetic interactions among our genes) cannot yet test link 4 cross-species; we record that as a coverage gap, not a failure (`data/raw/biogrid_ecoli_overrescue_interactions.tsv`).

## 8 Discussion

The honest negative on MCC is scientifically informative: with 159 positives among 1,107 genes, a per-fold threshold tuned on training scores is unstable, and the FBA rule’s discrete near-zero

| chain link                          | <i>S. cerevisiae</i>              | <i>E. coli</i>                    | <i>B. subtilis</i>                |
|-------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 1. missed essentials isozyme-backed | OR 16.2, $p = 3.8 \times 10^{-4}$ | OR 11.5, $p = 1.2 \times 10^{-5}$ | OR 22.7, $p = 2.5 \times 10^{-7}$ |
| 2. named over-rescued genes         | 19                                | 13                                | 15                                |
| 3. backups less abundant            | 17/28, $p = 0.038$                | 17/20, $p = 0.007$                | 38/71, $p = 0.29$ (neg.)          |
| 4. overexpression rescues (curated) | 4 pairs (BioGRID)                 | untested (no curation)            | untested                          |

Table 7: The isozyme over-rescue chain across three species. Link 3 abundances from PaxDb v4.2; link 4 from BioGRID 5.0.261 rescue-type interactions between model backup pairs.

knockout ratios are already near-optimal hard calls. The ranking result shows the graph network learns essentiality signal the rule does not carry—consistent with the over-rescue discovery, since the network sees GPR redundancy as a feature rather than a logical crutch. Isozyme over-rescue is actionable: the auditor flags exactly which viability predictions rest on unvalidated backup clauses, and the same audit runs on any SBML model.

## 8.1 Limitations

The gold set (1,107 genes) covers the model’s metabolic core but not every gene; essentiality is medium-specific and our claims inherit the Kennedy-Y7 condition for essentiality and minimal-glucose for strain design. The GCN is transductive: it exploits the full graph including unlabelled structure, which is legitimate here but should be re-checked under a fully inductive split for prospective use. The strain-design candidates are in-silico predictions; their value is that they are falsifiable with standard two-gene deletions.

## 8.2 Future work

Inductive (graph-split) evaluation; expression-conditioned flux bounds to sharpen both essentiality and design; applying the over-rescue auditor to bacterial and human genome-scale models; and experimental testing of  $\Delta\text{sdh2}\Delta\text{ach1}$ , which is a routine two-deletion construct.

# 9 Per-model fold analysis

The fifteen fold-level outcomes (Table 2) reward a close read. The FBA rule is bimodal across folds: when a fold’s essentials are concentrated in low-redundancy reactions its MCC approaches 0.68; folds rich in over-rescued genes drag it toward 0.32. The GCN’s fold variance tracks the same quantity with a lag, because its features include the knockout ratio itself; its advantage is concentrated in exactly the folds where the rule is weakest, which is the signature of genuinely complementary signal rather than redundant copying. Logistic regression traces the GCN at slightly lower amplitude, indicating that much of the recoverable signal is already linear in the eleven features, and that the graph adds value mainly through the redundancy structure (eq. (5)). The ensemble’s poor showing is entirely attributable to the sequence CNN’s collapse propagating through the averaged scores; we report it rather than silently dropping the combination, because the locked protocol fixed the ensemble’s definition before any fold was run.

## 10 Conclusion

A virtual yeast cell built on the consensus genome-scale model reproduces the published essentiality benchmark, fails to beat it on hard calls under a locked protocol (preserved negative), beats it decisively on ranking ( $\Delta\text{AUC} +0.135$ ), and yields a named, quantified, falsifiable failure mode—isozyme over-rescue—with a reusable auditor and 19 named genes. The failure mode replicates across *E. coli* and *B. subtilis*, its mechanism is expression-blindness combined with annotation transfer (PaxDb, BioGRID, KEGG evidence), and its one non-replication (*B. subtilis* abundances) is reported as an honest negative. For metabolic engineering, the full knockout scan validates the published quadruple succinate strain and nominates a simpler, faster-growing double knockout ( $\Delta\text{sdh2}\Delta\text{ach1}$ ) whose obligatory production exceeds the published strain at near-wild-type growth—a falsifiable prediction awaiting the bench.

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## A Reproducibility

Seeds: CV folds 20260924, bootstrap 7. All tests are hermetic; the SBML, labels and sequences ship in-repository. Environment: Python 3.10.12, COBRApy 0.32.1, PyTorch 2.14.0 (CPU), scikit-learn 1.7.2.

## B Feature definitions

Full per-gene feature table (1,143 genes  $\times$  11 features) at `results/gene_features_partial.csv`.

## C Strain-design tables

Full scan outputs: `results/strain-design-*.csv`.

## D Per-gene essentiality scores

Table 8: All 159 observed-essential genes, ranked by the full-data GCN score. FBA-missed genes (rank column shows the GCN’s recovery ordering).

| rank | gene    | name  | GCN   | FBA    | LR    |
|------|---------|-------|-------|--------|-------|
| 1    | YIL083C | CAB2  | 1.000 | 0.100  | 0.935 |
| 2    | YKL004W | AUR1  | 0.999 | -0.182 | 0.545 |
| 3    | YDR367W | KEI1  | 0.999 | -0.182 | 0.545 |
| 4    | YNL256W | FOL1  | 0.999 | 0.100  | 0.964 |
| 5    | YDL055C | PSA1  | 0.998 | 0.100  | 0.952 |
| 6    | YDR454C | GUK1  | 0.998 | 0.100  | 0.941 |
| 7    | YOR176W | HEM15 | 0.998 | 0.100  | 0.962 |
| 8    | YNR043W | MVD1  | 0.997 | 0.100  | 0.964 |
| 9    | YPR183W | DPM1  | 0.997 | 0.100  | 0.928 |
| 10   | YDL103C | QRI1  | 0.996 | 0.100  | 0.956 |
| 11   | YDR047W | HEM12 | 0.996 | 0.100  | 0.965 |
| 12   | YOR278W | HEM4  | 0.996 | 0.100  | 0.965 |
| 13   | YJL167W | ERG20 | 0.995 | 0.100  | 0.964 |

|    |         |         |       |        |       |
|----|---------|---------|-------|--------|-------|
| 14 | YOR168W | GLN4    | 0.995 | 0.100  | 0.942 |
| 15 | YGR267C | FOL2    | 0.995 | 0.100  | 0.946 |
| 16 | YDL205C | HEM3    | 0.995 | 0.100  | 0.965 |
| 17 | YEL058W | PCM1    | 0.995 | 0.100  | 0.961 |
| 18 | YPL117C | IDI1    | 0.995 | 0.100  | 0.965 |
| 19 | YPR033C | HTS1    | 0.995 | 0.100  | 0.947 |
| 20 | YKL024C | URA6    | 0.994 | 0.100  | 0.909 |
| 21 | YFL017C | GNA1    | 0.994 | 0.100  | 0.945 |
| 22 | YNL247W | YNL247W | 0.994 | 0.100  | 0.956 |
| 23 | YGL040C | HEM2    | 0.994 | 0.100  | 0.965 |
| 24 | YMR208W | ERG12   | 0.994 | 0.100  | 0.894 |
| 25 | YGR185C | TYS1    | 0.994 | 0.100  | 0.956 |
| 26 | YOR074C | CDC21   | 0.994 | 0.100  | 0.956 |
| 27 | YDL045C | FAD1    | 0.993 | 0.100  | 0.952 |
| 28 | YDR044W | HEM13   | 0.993 | 0.100  | 0.966 |
| 29 | YPL160W | CDC60   | 0.992 | 0.100  | 0.959 |
| 30 | YGR264C | MES1    | 0.991 | 0.100  | 0.952 |
| 31 | YDR196C | CAB5    | 0.991 | 0.100  | 0.956 |
| 32 | YGR094W | VAS1    | 0.990 | 0.100  | 0.951 |
| 33 | YBL076C | ILS1    | 0.990 | 0.100  | 0.959 |
| 34 | YMR220W | ERG8    | 0.990 | 0.100  | 0.965 |
| 35 | YDR037W | KRS1    | 0.990 | 0.100  | 0.960 |
| 36 | YDR341C | YDR341C | 0.989 | 0.100  | 0.960 |
| 37 | YLL018C | DPS1    | 0.989 | 0.100  | 0.956 |
| 38 | YOL097C | WRS1    | 0.989 | 0.100  | 0.960 |
| 39 | YDR531W | CAB1    | 0.988 | 0.100  | 0.965 |
| 40 | YHR019C | DED81   | 0.988 | 0.100  | 0.960 |
| 41 | YER003C | PMI40   | 0.986 | 0.100  | 0.953 |
| 42 | YHR020W | YHR020W | 0.986 | 0.100  | 0.965 |
| 43 | YDR373W | FRQ1    | 0.985 | -0.182 | 0.524 |
| 44 | YIL078W | THS1    | 0.984 | 0.100  | 0.962 |
| 45 | YOR095C | RKI1    | 0.983 | 0.100  | 0.948 |
| 46 | YNL267W | PIK1    | 0.983 | -0.182 | 0.484 |
| 47 | YOR335C | ALA1    | 0.982 | 0.100  | 0.959 |
| 48 | YHR074W | QNS1    | 0.982 | 0.100  | 0.871 |
| 49 | YGL245W | GUS1    | 0.981 | 0.100  | 0.909 |
| 50 | YDR208W | MSS4    | 0.979 | -0.182 | 0.497 |
| 51 | YFL045C | SEC53   | 0.976 | 0.100  | 0.963 |
| 52 | YOR236W | DFR1    | 0.975 | 0.100  | 0.939 |
| 53 | YMR113W | FOL3    | 0.975 | 0.100  | 0.906 |
| 54 | YLR060W | FRS1    | 0.975 | 0.100  | 0.958 |
| 55 | YFL022C | FRS2    | 0.975 | 0.100  | 0.958 |
| 56 | YGL047W | ALG13   | 0.972 | -0.182 | 0.491 |
| 57 | YBR070C | ALG14   | 0.972 | -0.182 | 0.491 |
| 58 | YKL088W | CAB3    | 0.971 | 0.100  | 0.959 |

|     |         |       |       |        |       |
|-----|---------|-------|-------|--------|-------|
| 59  | YER043C | SAH1  | 0.969 | 0.100  | 0.949 |
| 60  | YGL055W | OLE1  | 0.967 | 0.100  | 0.957 |
| 61  | YDR437W | GPI19 | 0.963 | -0.182 | 0.420 |
| 62  | YNL038W | GPI15 | 0.963 | -0.182 | 0.420 |
| 63  | YPL175W | SPT14 | 0.963 | -0.182 | 0.420 |
| 64  | YGR216C | GPI1  | 0.963 | -0.182 | 0.420 |
| 65  | YPL076W | GPI2  | 0.963 | -0.182 | 0.420 |
| 66  | YPL252C | YAH1  | 0.960 | -0.182 | 0.250 |
| 67  | YDR376W | ARH1  | 0.960 | -0.182 | 0.250 |
| 68  | YNR016C | ACC1  | 0.956 | 0.100  | 0.936 |
| 69  | YML126C | ERG13 | 0.954 | 0.100  | 0.950 |
| 70  | YIL048W | NEO1  | 0.951 | -0.182 | 0.296 |
| 71  | YPL028W | ERG10 | 0.950 | 0.100  | 0.918 |
| 72  | YJL101C | GSH1  | 0.934 | 0.100  | 0.884 |
| 73  | YPR165W | RHO1  | 0.932 | 0.100  | 0.792 |
| 74  | YDL141W | BPL1  | 0.931 | 0.100  | 0.886 |
| 75  | YDR232W | HEM1  | 0.930 | 0.100  | 0.958 |
| 76  | YBR029C | CDS1  | 0.927 | 0.100  | 0.949 |
| 77  | YBR002C | RER2  | 0.919 | -0.182 | 0.301 |
| 78  | YBR243C | ALG7  | 0.919 | -0.182 | 0.435 |
| 79  | YJR013W | GPI14 | 0.916 | -0.182 | 0.441 |
| 80  | YCL052C | PBN1  | 0.916 | -0.182 | 0.441 |
| 81  | YOL049W | GSH2  | 0.916 | 0.100  | 0.933 |
| 82  | YMR296C | LCB1  | 0.914 | -0.182 | 0.429 |
| 83  | YDR062W | LCB2  | 0.914 | -0.182 | 0.429 |
| 84  | YGL225W | VRG4  | 0.906 | -0.182 | 0.412 |
| 85  | YPR113W | PIS1  | 0.901 | 0.100  | 0.958 |
| 86  | YBR192W | RIM2  | 0.895 | -0.182 | 0.225 |
| 87  | YDL166C | FAP7  | 0.890 | -0.182 | 0.514 |
| 88  | YOL130W | ALR1  | 0.874 | 0.100  | 0.956 |
| 89  | YLR305C | STT4  | 0.872 | -0.182 | 0.342 |
| 90  | YKL104C | GFA1  | 0.870 | 0.100  | 0.864 |
| 91  | YJL091C | GWT1  | 0.870 | -0.182 | 0.514 |
| 92  | YER023W | PRO3  | 0.867 | -0.179 | 0.569 |
| 93  | YGR060W | ERG25 | 0.856 | -0.182 | 0.493 |
| 94  | YKL152C | GPM1  | 0.848 | -0.182 | 0.474 |
| 95  | YKL019W | RAM2  | 0.831 | -0.182 | 0.536 |
| 96  | YBR110W | ALG1  | 0.831 | -0.182 | 0.469 |
| 97  | YOR143C | THI80 | 0.830 | -0.182 | 0.557 |
| 98  | YGR277C | CAB4  | 0.826 | -0.182 | 0.582 |
| 99  | YHR068W | DYS1  | 0.807 | -0.182 | 0.332 |
| 100 | YCR012W | PGK1  | 0.797 | -0.174 | 0.546 |
| 101 | YLR153C | ACS2  | 0.776 | -0.182 | 0.335 |
| 102 | YGL142C | GPI10 | 0.773 | -0.182 | 0.489 |
| 103 | YHR042W | NCP1  | 0.769 | -0.182 | 0.466 |



|     |         |       |       |        |       |
|-----|---------|-------|-------|--------|-------|
| 104 | YHR007C | ERG11 | 0.769 | -0.182 | 0.466 |
| 105 | YPL231W | FAS2  | 0.764 | -0.182 | 0.418 |
| 106 | YKL182W | FAS1  | 0.764 | -0.182 | 0.418 |
| 107 | YJR016C | ILV3  | 0.764 | -0.048 | 0.850 |
| 108 | YDL120W | YFH1  | 0.760 | -0.182 | 0.544 |
| 109 | YHR190W | ERG9  | 0.758 | -0.182 | 0.544 |
| 110 | YBR265W | TSC10 | 0.752 | -0.182 | 0.484 |
| 111 | YLR100W | ERG27 | 0.750 | -0.182 | 0.535 |
| 112 | YBR004C | GPI18 | 0.748 | -0.182 | 0.464 |
| 113 | YKL165C | MCD4  | 0.742 | -0.182 | 0.534 |
| 114 | YJL005W | CYR1  | 0.728 | -0.182 | 0.480 |
| 115 | YOR149C | SMP3  | 0.707 | -0.182 | 0.469 |
| 116 | YMR298W | LIP1  | 0.705 | -0.182 | 0.462 |
| 117 | YMR013C | SEC59 | 0.699 | -0.182 | 0.479 |
| 118 | YKL060C | FBA1  | 0.696 | -0.182 | 0.428 |
| 119 | YGL001C | ERG26 | 0.686 | -0.182 | 0.550 |
| 120 | YKL035W | UGP1  | 0.677 | -0.182 | 0.314 |
| 121 | YBR153W | RIB7  | 0.667 | -0.182 | 0.549 |
| 122 | YBR256C | RIB5  | 0.657 | -0.182 | 0.539 |
| 123 | YDR236C | FMN1  | 0.649 | -0.182 | 0.476 |
| 124 | YAL038W | CDC19 | 0.636 | -0.182 | 0.247 |
| 125 | YJR057W | CDC8  | 0.634 | -0.182 | 0.534 |
| 126 | YLR355C | ILV5  | 0.629 | -0.048 | 0.852 |
| 127 | YMR108W | ILV2  | 0.626 | -0.048 | 0.658 |
| 128 | YMR301C | ATM1  | 0.617 | -0.182 | 0.515 |
| 129 | YBL030C | PET9  | 0.612 | -0.182 | 0.102 |
| 130 | YMR281W | GPI12 | 0.611 | -0.182 | 0.554 |
| 131 | YBR011C | IPP1  | 0.607 | -0.182 | 0.559 |
| 132 | YOL066C | RIB2  | 0.603 | -0.182 | 0.554 |
| 133 | YGR191W | HIP1  | 0.602 | -0.182 | 0.341 |
| 134 | YML125C | PGA3  | 0.596 | -0.182 | 0.352 |
| 135 | YJL026W | RNR2  | 0.585 | -0.182 | 0.040 |
| 136 | YGL008C | PMA1  | 0.579 | -0.182 | 0.244 |
| 137 | YGL065C | ALG2  | 0.577 | -0.182 | 0.550 |
| 138 | YHR072W | ERG7  | 0.548 | -0.182 | 0.539 |
| 139 | YGR046W | TAM41 | 0.546 | -0.182 | 0.544 |
| 140 | YDR050C | TPI1  | 0.542 | -0.182 | 0.499 |
| 141 | YGR175C | ERG1  | 0.540 | -0.182 | 0.560 |
| 142 | YJL097W | PHS1  | 0.525 | -0.181 | 0.565 |
| 143 | YBR196C | PGI1  | 0.522 | -0.182 | 0.458 |
| 144 | YBR252W | DUT1  | 0.493 | -0.182 | 0.385 |
| 145 | YGL169W | SUA5  | 0.490 | -0.182 | 0.479 |
| 146 | YBR038W | CHS2  | 0.487 | -0.182 | 0.210 |
| 147 | YLL031C | GPI13 | 0.485 | -0.182 | 0.399 |
| 148 | YDR302W | GPI11 | 0.485 | -0.182 | 0.399 |

|     |         |        |       |        |       |
|-----|---------|--------|-------|--------|-------|
| 149 | YGR065C | VHT1   | 0.446 | -0.182 | 0.505 |
| 150 | YDR353W | TRR1   | 0.442 | -0.182 | 0.255 |
| 151 | YDL015C | TSC13  | 0.436 | -0.181 | 0.506 |
| 152 | YHR128W | FUR1   | 0.410 | -0.182 | 0.405 |
| 153 | YDL004W | ATP16  | 0.404 | -0.006 | 0.447 |
| 154 | YDR023W | SES1   | 0.369 | -0.182 | 0.378 |
| 155 | YKL141W | SDH3   | 0.359 | -0.177 | 0.142 |
| 156 | YDR487C | RIB3   | 0.327 | -0.182 | 0.464 |
| 157 | YNL036W | NCE103 | 0.315 | -0.182 | 0.555 |
| 158 | YPR035W | GLN1   | 0.191 | -0.182 | 0.285 |
| 159 | YLR359W | ADE13  | 0.162 | -0.182 | 0.510 |

## E Full tool output: all benchmark genes

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 1    | YIL083C | CAB2    | 1         | 1.000 | 0.100  | 0.935 |
| 2    | YDR072C | IPT1    | 0         | 0.999 | -0.182 | 0.563 |
| 3    | YKL004W | AUR1    | 1         | 0.999 | -0.182 | 0.545 |
| 4    | YDR367W | KEI1    | 1         | 0.999 | -0.182 | 0.545 |
| 5    | YNL256W | FOL1    | 1         | 0.999 | 0.100  | 0.964 |
| 6    | YDL055C | PSA1    | 1         | 0.998 | 0.100  | 0.952 |
| 7    | YDR454C | GUK1    | 1         | 0.998 | 0.100  | 0.941 |
| 8    | YOR176W | HEM15   | 1         | 0.998 | 0.100  | 0.962 |
| 9    | YPL172C | COX10   | 0         | 0.997 | 0.100  | 0.963 |
| 10   | YNR043W | MVD1    | 1         | 0.997 | 0.100  | 0.964 |
| 11   | YPR183W | DPM1    | 1         | 0.997 | 0.100  | 0.928 |
| 12   | YDL103C | QRI1    | 1         | 0.996 | 0.100  | 0.956 |
| 13   | YDR047W | HEM12   | 1         | 0.996 | 0.100  | 0.965 |
| 14   | YOR278W | HEM4    | 1         | 0.996 | 0.100  | 0.965 |
| 15   | YJL167W | ERG20   | 1         | 0.995 | 0.100  | 0.964 |
| 16   | YOR168W | GLN4    | 1         | 0.995 | 0.100  | 0.942 |
| 17   | YGR267C | FOL2    | 1         | 0.995 | 0.100  | 0.946 |
| 18   | YDL205C | HEM3    | 1         | 0.995 | 0.100  | 0.965 |
| 19   | YEL058W | PCM1    | 1         | 0.995 | 0.100  | 0.961 |
| 20   | YPL117C | IDI1    | 1         | 0.995 | 0.100  | 0.965 |
| 21   | YPR033C | HTS1    | 1         | 0.995 | 0.100  | 0.947 |
| 22   | YKL024C | URA6    | 1         | 0.994 | 0.100  | 0.909 |
| 23   | YFL017C | GNA1    | 1         | 0.994 | 0.100  | 0.945 |
| 24   | YNL247W | YNL247W | 1         | 0.994 | 0.100  | 0.956 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 25   | YGL040C | HEM2    | 1         | 0.994 | 0.100  | 0.965 |
| 26   | YMR208W | ERG12   | 1         | 0.994 | 0.100  | 0.894 |
| 27   | YGR185C | TYS1    | 1         | 0.994 | 0.100  | 0.956 |
| 28   | YOR074C | CDC21   | 1         | 0.994 | 0.100  | 0.956 |
| 29   | YDL045C | FAD1    | 1         | 0.993 | 0.100  | 0.952 |
| 30   | YDR044W | HEM13   | 1         | 0.993 | 0.100  | 0.966 |
| 31   | YPL160W | CDC60   | 1         | 0.992 | 0.100  | 0.959 |
| 32   | YGR264C | MES1    | 1         | 0.991 | 0.100  | 0.952 |
| 33   | YDR196C | CAB5    | 1         | 0.991 | 0.100  | 0.956 |
| 34   | YGR094W | VAS1    | 1         | 0.990 | 0.100  | 0.951 |
| 35   | YBL076C | ILS1    | 1         | 0.990 | 0.100  | 0.959 |
| 36   | YMR220W | ERG8    | 1         | 0.990 | 0.100  | 0.965 |
| 37   | YDR037W | KRS1    | 1         | 0.990 | 0.100  | 0.960 |
| 38   | YDR341C | YDR341C | 1         | 0.989 | 0.100  | 0.960 |
| 39   | YLL018C | DPS1    | 1         | 0.989 | 0.100  | 0.956 |
| 40   | YOL097C | WRS1    | 1         | 0.989 | 0.100  | 0.960 |
| 41   | YDR531W | CAB1    | 1         | 0.988 | 0.100  | 0.965 |
| 42   | YHR019C | DED81   | 1         | 0.988 | 0.100  | 0.960 |
| 43   | YER014W | HEM14   | 0         | 0.986 | 0.100  | 0.966 |
| 44   | YER003C | PMI40   | 1         | 0.986 | 0.100  | 0.953 |
| 45   | YHR020W | YHR020W | 1         | 0.986 | 0.100  | 0.965 |
| 46   | YDR373W | FRQ1    | 1         | 0.985 | -0.182 | 0.524 |
| 47   | YIL078W | THS1    | 1         | 0.984 | 0.100  | 0.962 |
| 48   | YOR095C | RKI1    | 1         | 0.983 | 0.100  | 0.948 |
| 49   | YNL267W | PIK1    | 1         | 0.983 | -0.182 | 0.484 |
| 50   | YOR335C | ALA1    | 1         | 0.982 | 0.100  | 0.959 |
| 51   | YHR074W | QNS1    | 1         | 0.982 | 0.100  | 0.871 |
| 52   | YGL245W | GUS1    | 1         | 0.981 | 0.100  | 0.909 |
| 53   | YDR208W | MSS4    | 1         | 0.979 | -0.182 | 0.497 |
| 54   | YFL045C | SEC53   | 1         | 0.976 | 0.100  | 0.963 |
| 55   | YOR236W | DFR1    | 1         | 0.975 | 0.100  | 0.939 |
| 56   | YMR113W | FOL3    | 1         | 0.975 | 0.100  | 0.906 |
| 57   | YLR060W | FRS1    | 1         | 0.975 | 0.100  | 0.958 |
| 58   | YFL022C | FRS2    | 1         | 0.975 | 0.100  | 0.958 |
| 59   | YGL047W | ALG13   | 1         | 0.972 | -0.182 | 0.491 |
| 60   | YBR070C | ALG14   | 1         | 0.972 | -0.182 | 0.491 |
| 61   | YKR072C | SIS2    | 0         | 0.971 | 0.100  | 0.959 |
| 62   | YKL088W | CAB3    | 1         | 0.971 | 0.100  | 0.959 |
| 63   | YOR054C | VHS3    | 0         | 0.971 | 0.100  | 0.959 |
| 64   | YER043C | SAH1    | 1         | 0.969 | 0.100  | 0.949 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name  | essential | GCN   | FBA    | LR    |
|------|-----------|-------|-----------|-------|--------|-------|
| 65   | YGL055W   | OLE1  | 1         | 0.967 | 0.100  | 0.957 |
| 66   | YPL096C-A | ERI1  | 0         | 0.963 | -0.182 | 0.420 |
| 67   | YPL175W   | SPT14 | 1         | 0.963 | -0.182 | 0.420 |
| 68   | YPL076W   | GPI2  | 1         | 0.963 | -0.182 | 0.420 |
| 69   | YDR437W   | GPI19 | 1         | 0.963 | -0.182 | 0.420 |
| 70   | YNL038W   | GPI15 | 1         | 0.963 | -0.182 | 0.420 |
| 71   | YGR216C   | GPI1  | 1         | 0.963 | -0.182 | 0.420 |
| 72   | YPL252C   | YAH1  | 1         | 0.960 | -0.182 | 0.250 |
| 73   | YDR376W   | ARH1  | 1         | 0.960 | -0.182 | 0.250 |
| 74   | YNR016C   | ACC1  | 1         | 0.956 | 0.100  | 0.936 |
| 75   | YML126C   | ERG13 | 1         | 0.954 | 0.100  | 0.950 |
| 76   | YIL048W   | NEO1  | 1         | 0.951 | -0.182 | 0.296 |
| 77   | YPL028W   | ERG10 | 1         | 0.950 | 0.100  | 0.918 |
| 78   | YER026C   | CHO1  | 0         | 0.938 | 0.100  | 0.961 |
| 79   | YJL101C   | GSH1  | 1         | 0.934 | 0.100  | 0.884 |
| 80   | YPR165W   | RHO1  | 1         | 0.932 | 0.100  | 0.792 |
| 81   | YDL141W   | BPL1  | 1         | 0.931 | 0.100  | 0.886 |
| 82   | YDR232W   | HEM1  | 1         | 0.930 | 0.100  | 0.958 |
| 83   | YBR029C   | CDS1  | 1         | 0.927 | 0.100  | 0.949 |
| 84   | YMR101C   | SRT1  | 0         | 0.919 | -0.182 | 0.301 |
| 85   | YBR002C   | RER2  | 1         | 0.919 | -0.182 | 0.301 |
| 86   | YBR243C   | ALG7  | 1         | 0.919 | -0.182 | 0.435 |
| 87   | YJR013W   | GPI14 | 1         | 0.916 | -0.182 | 0.441 |
| 88   | YCL052C   | PBN1  | 1         | 0.916 | -0.182 | 0.441 |
| 89   | YOL049W   | GSH2  | 1         | 0.916 | 0.100  | 0.933 |
| 90   | YPL268W   | PLC1  | 0         | 0.915 | -0.182 | 0.507 |
| 91   | YDR062W   | LCB2  | 1         | 0.914 | -0.182 | 0.429 |
| 92   | YMR296C   | LCB1  | 1         | 0.914 | -0.182 | 0.429 |
| 93   | YGR038W   | ORM1  | 0         | 0.914 | -0.182 | 0.429 |
| 94   | YBR058C-A | TSC3  | 0         | 0.914 | -0.182 | 0.429 |
| 95   | YLR350W   | ORM2  | 0         | 0.914 | -0.182 | 0.429 |
| 96   | YPL227C   | ALG5  | 0         | 0.909 | -0.182 | 0.440 |
| 97   | YGL225W   | VRG4  | 1         | 0.906 | -0.182 | 0.412 |
| 98   | YMR217W   | GUA1  | 0         | 0.901 | 0.100  | 0.882 |
| 99   | YPR113W   | PIS1  | 1         | 0.901 | 0.100  | 0.958 |
| 100  | YBR192W   | RIM2  | 1         | 0.895 | -0.182 | 0.225 |
| 101  | YDL166C   | FAP7  | 1         | 0.890 | -0.182 | 0.514 |
| 102  | YEL004W   | YEA4  | 0         | 0.882 | -0.182 | 0.470 |
| 103  | YOL130W   | ALR1  | 1         | 0.874 | 0.100  | 0.956 |
| 104  | YLR305C   | STT4  | 1         | 0.872 | -0.182 | 0.342 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 105  | YKL104C | GFA1  | 1         | 0.870 | 0.100  | 0.864 |
| 106  | YJL091C | GWT1  | 1         | 0.870 | -0.182 | 0.514 |
| 107  | YER023W | PRO3  | 1         | 0.867 | -0.179 | 0.569 |
| 108  | YPL069C | BTS1  | 0         | 0.863 | -0.182 | 0.534 |
| 109  | YGR060W | ERG25 | 1         | 0.856 | -0.182 | 0.493 |
| 110  | YKL152C | GPM1  | 1         | 0.848 | -0.182 | 0.474 |
| 111  | YHR144C | DCD1  | 0         | 0.839 | -0.182 | 0.542 |
| 112  | YKL019W | RAM2  | 1         | 0.831 | -0.182 | 0.536 |
| 113  | YDL090C | RAM1  | 0         | 0.831 | -0.182 | 0.536 |
| 114  | YBR110W | ALG1  | 1         | 0.831 | -0.182 | 0.469 |
| 115  | YOR143C | THI80 | 1         | 0.830 | -0.182 | 0.557 |
| 116  | YGR277C | CAB4  | 1         | 0.826 | -0.182 | 0.582 |
| 117  | YHR068W | DYS1  | 1         | 0.807 | -0.182 | 0.332 |
| 118  | YCR012W | PGK1  | 1         | 0.797 | -0.174 | 0.546 |
| 119  | YGL009C | LEU1  | 0         | 0.789 | -0.038 | 0.860 |
| 120  | YPL214C | THI6  | 0         | 0.789 | -0.182 | 0.536 |
| 121  | YFL058W | THI5  | 0         | 0.778 | -0.182 | 0.534 |
| 122  | YLR153C | ACS2  | 1         | 0.776 | -0.182 | 0.335 |
| 123  | YKR069W | MET1  | 0         | 0.774 | -0.182 | 0.464 |
| 124  | YNL275W | BOR1  | 0         | 0.773 | 0.100  | 0.959 |
| 125  | YGL142C | GPI10 | 1         | 0.773 | -0.182 | 0.489 |
| 126  | YHR042W | NCP1  | 1         | 0.769 | -0.182 | 0.466 |
| 127  | YHR007C | ERG11 | 1         | 0.769 | -0.182 | 0.466 |
| 128  | YPL231W | FAS2  | 1         | 0.764 | -0.182 | 0.418 |
| 129  | YKL182W | FAS1  | 1         | 0.764 | -0.182 | 0.418 |
| 130  | YJR016C | ILV3  | 1         | 0.764 | -0.048 | 0.850 |
| 131  | YDL120W | YFH1  | 1         | 0.760 | -0.182 | 0.544 |
| 132  | YHR190W | ERG9  | 1         | 0.758 | -0.182 | 0.544 |
| 133  | YER141W | COX15 | 0         | 0.756 | -0.182 | 0.442 |
| 134  | YER061C | CEM1  | 0         | 0.755 | -0.182 | 0.523 |
| 135  | YBR265W | TSC10 | 1         | 0.752 | -0.182 | 0.484 |
| 136  | YLR100W | ERG27 | 1         | 0.750 | -0.182 | 0.535 |
| 137  | YBR004C | GPI18 | 1         | 0.748 | -0.182 | 0.464 |
| 138  | YDL210W | UGA4  | 0         | 0.747 | -0.182 | 0.483 |
| 139  | YKL165C | MCD4  | 1         | 0.742 | -0.182 | 0.534 |
| 140  | YIR027C | DAL1  | 0         | 0.742 | -0.182 | 0.544 |
| 141  | YLR056W | ERG3  | 0         | 0.738 | -0.182 | 0.539 |
| 142  | YDL198C | GGC1  | 0         | 0.733 | -0.182 | 0.406 |
| 143  | YJL005W | CYR1  | 1         | 0.728 | -0.182 | 0.480 |
| 144  | YLR133W | CKI1  | 0         | 0.720 | -0.182 | 0.517 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 145  | YLR011W | LOT6    | 0         | 0.714 | -0.182 | 0.530 |
| 146  | YDR503C | LPP1    | 0         | 0.713 | -0.182 | 0.566 |
| 147  | YBR111C | YSA1    | 0         | 0.711 | -0.182 | 0.549 |
| 148  | YOR149C | SMP3    | 1         | 0.707 | -0.182 | 0.469 |
| 149  | YNR030W | ALG12   | 0         | 0.707 | -0.182 | 0.469 |
| 150  | YMR298W | LIP1    | 1         | 0.705 | -0.182 | 0.462 |
| 151  | YHL003C | LAG1    | 0         | 0.705 | -0.182 | 0.462 |
| 152  | YKL008C | LAC1    | 0         | 0.705 | -0.182 | 0.462 |
| 153  | YNL219C | ALG9    | 0         | 0.704 | -0.182 | 0.470 |
| 154  | YGR170W | PSD2    | 0         | 0.703 | -0.182 | 0.526 |
| 155  | YNL045W | LAP2    | 0         | 0.702 | -0.182 | 0.521 |
| 156  | YMR013C | SEC59   | 1         | 0.699 | -0.182 | 0.479 |
| 157  | YKL060C | FBA1    | 1         | 0.696 | -0.182 | 0.428 |
| 158  | YGR036C | CAX4    | 0         | 0.695 | -0.182 | 0.479 |
| 159  | YOR002W | ALG6    | 0         | 0.695 | -0.182 | 0.479 |
| 160  | YOR067C | ALG8    | 0         | 0.695 | -0.182 | 0.479 |
| 161  | YGR227W | DIE2    | 0         | 0.695 | -0.182 | 0.479 |
| 162  | YCL018W | LEU2    | 0         | 0.690 | -0.038 | 0.867 |
| 163  | YGL001C | ERG26   | 1         | 0.686 | -0.182 | 0.550 |
| 164  | YBR229C | ROT2    | 0         | 0.678 | -0.182 | 0.545 |
| 165  | YKL035W | UGP1    | 1         | 0.677 | -0.182 | 0.314 |
| 166  | YJR139C | HOM6    | 0         | 0.668 | -0.113 | 0.729 |
| 167  | YBR153W | RIB7    | 1         | 0.667 | -0.182 | 0.549 |
| 168  | YER019W | ISC1    | 0         | 0.666 | -0.182 | 0.544 |
| 169  | YGL038C | OCH1    | 0         | 0.663 | -0.182 | 0.539 |
| 170  | YBR256C | RIB5    | 1         | 0.657 | -0.182 | 0.539 |
| 171  | YER052C | HOM3    | 0         | 0.651 | -0.113 | 0.697 |
| 172  | YMR015C | ERG5    | 0         | 0.649 | -0.182 | 0.554 |
| 173  | YDR236C | FMN1    | 1         | 0.649 | -0.182 | 0.476 |
| 174  | YOR283W | YOR283W | 0         | 0.639 | -0.182 | 0.374 |
| 175  | YNL280C | ERG24   | 0         | 0.636 | -0.182 | 0.544 |
| 176  | YOR347C | PYK2    | 0         | 0.636 | -0.182 | 0.247 |
| 177  | YAL038W | CDC19   | 1         | 0.636 | -0.182 | 0.247 |
| 178  | YGR286C | BIO2    | 0         | 0.634 | -0.182 | 0.544 |
| 179  | YJR057W | CDC8    | 1         | 0.634 | -0.182 | 0.534 |
| 180  | YOL122C | SMF1    | 0         | 0.630 | 0.100  | 0.958 |
| 181  | YLR355C | ILV5    | 1         | 0.629 | -0.048 | 0.852 |
| 182  | YPL110C | GDE1    | 0         | 0.628 | -0.182 | 0.514 |
| 183  | YBR003W | COQ1    | 0         | 0.627 | -0.182 | 0.554 |
| 184  | YNR041C | COQ2    | 0         | 0.627 | -0.182 | 0.554 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name  | essential | GCN   | FBA    | LR    |
|------|-----------|-------|-----------|-------|--------|-------|
| 185  | YMR108W   | ILV2  | 1         | 0.626 | -0.048 | 0.658 |
| 186  | YLR142W   | PUT1  | 0         | 0.626 | -0.182 | 0.549 |
| 187  | YIL099W   | SGA1  | 0         | 0.624 | -0.182 | 0.544 |
| 188  | YOR221C   | MCT1  | 0         | 0.623 | -0.182 | 0.526 |
| 189  | YHR063C   | PAN5  | 0         | 0.623 | -0.182 | 0.554 |
| 190  | YPL057C   | SUR1  | 0         | 0.622 | -0.182 | 0.278 |
| 191  | YBR161W   | CSH1  | 0         | 0.622 | -0.182 | 0.278 |
| 192  | YMR301C   | ATM1  | 1         | 0.617 | -0.182 | 0.515 |
| 193  | YLL061W   | MMP1  | 0         | 0.617 | -0.182 | 0.515 |
| 194  | YIL134W   | FLX1  | 0         | 0.617 | -0.182 | 0.515 |
| 195  | YLL048C   | YBT1  | 0         | 0.617 | -0.182 | 0.515 |
| 196  | YCR075C   | ERS1  | 0         | 0.617 | -0.182 | 0.515 |
| 197  | YCR011C   | ADP1  | 0         | 0.617 | -0.182 | 0.515 |
| 198  | YBR295W   | PCA1  | 0         | 0.617 | -0.182 | 0.515 |
| 199  | YMR056C   | AAC1  | 0         | 0.612 | -0.182 | 0.102 |
| 200  | YBR085W   | AAC3  | 0         | 0.612 | -0.182 | 0.102 |
| 201  | YBL030C   | PET9  | 1         | 0.612 | -0.182 | 0.102 |
| 202  | YIL145C   | PAN6  | 0         | 0.612 | -0.182 | 0.534 |
| 203  | YGL154C   | LYS5  | 0         | 0.612 | -0.052 | 0.832 |
| 204  | YBR115C   | LYS2  | 0         | 0.612 | -0.052 | 0.832 |
| 205  | YMR281W   | GPI12 | 1         | 0.611 | -0.182 | 0.554 |
| 206  | YGR202C   | PCT1  | 0         | 0.608 | -0.182 | 0.504 |
| 207  | YGR154C   | GTO1  | 0         | 0.607 | -0.182 | 0.559 |
| 208  | YBR046C   | ZTA1  | 0         | 0.607 | -0.182 | 0.559 |
| 209  | YPL097W   | MSY1  | 0         | 0.607 | -0.182 | 0.559 |
| 210  | YPR047W   | MSF1  | 0         | 0.607 | -0.182 | 0.559 |
| 211  | YGL067W   | NPY1  | 0         | 0.607 | -0.182 | 0.559 |
| 212  | YDL236W   | PHO13 | 0         | 0.607 | -0.182 | 0.559 |
| 213  | YJR153W   | PGU1  | 0         | 0.607 | -0.182 | 0.559 |
| 214  | YBR222C   | PCS60 | 0         | 0.607 | -0.182 | 0.559 |
| 215  | YDR268W   | MSW1  | 0         | 0.607 | -0.182 | 0.559 |
| 216  | YGL012W   | ERG4  | 0         | 0.607 | -0.182 | 0.559 |
| 217  | YGL156W   | AMS1  | 0         | 0.607 | -0.182 | 0.559 |
| 218  | YBR011C   | IPP1  | 1         | 0.607 | -0.182 | 0.559 |
| 219  | YLR151C   | PCD1  | 0         | 0.607 | -0.182 | 0.559 |
| 220  | YOL066C   | RIB2  | 1         | 0.603 | -0.182 | 0.554 |
| 221  | YBR126C   | TPS1  | 0         | 0.603 | 0.100  | 0.850 |
| 222  | YDR074W   | TPS2  | 0         | 0.603 | 0.100  | 0.850 |
| 223  | YEL017C-A | PMP2  | 0         | 0.603 | -0.182 | 0.239 |
| 224  | YCR024C-A | PMP1  | 0         | 0.603 | -0.182 | 0.239 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 225  | YGR191W | HIP1  | 1         | 0.602 | -0.182 | 0.341 |
| 226  | YDR305C | HNT2  | 0         | 0.597 | -0.182 | 0.555 |
| 227  | YIL043C | CBR1  | 0         | 0.596 | -0.182 | 0.352 |
| 228  | YML125C | PGA3  | 1         | 0.596 | -0.182 | 0.352 |
| 229  | YAL026C | DRS2  | 0         | 0.595 | -0.182 | 0.346 |
| 230  | YNL129W | NRK1  | 0         | 0.593 | -0.182 | 0.544 |
| 231  | YIR034C | LYS1  | 0         | 0.587 | -0.052 | 0.745 |
| 232  | YGR180C | RNR4  | 0         | 0.585 | -0.182 | 0.040 |
| 233  | YER070W | RNR1  | 0         | 0.585 | -0.182 | 0.040 |
| 234  | YJL026W | RNR2  | 1         | 0.585 | -0.182 | 0.040 |
| 235  | YIL066C | RNR3  | 0         | 0.585 | -0.182 | 0.040 |
| 236  | YGL008C | PMA1  | 1         | 0.579 | -0.182 | 0.244 |
| 237  | YPL036W | PMA2  | 0         | 0.579 | -0.182 | 0.244 |
| 238  | YKR067W | GPT2  | 0         | 0.579 | 0.100  | 0.934 |
| 239  | YER087W | AIM10 | 0         | 0.579 | -0.182 | 0.554 |
| 240  | YNR057C | BIO4  | 0         | 0.578 | -0.182 | 0.549 |
| 241  | YPR118W | MRI1  | 0         | 0.578 | -0.182 | 0.549 |
| 242  | YGL065C | ALG2  | 1         | 0.577 | -0.182 | 0.550 |
| 243  | YNL048W | ALG11 | 0         | 0.577 | -0.182 | 0.550 |
| 244  | YDR530C | APA2  | 0         | 0.574 | -0.182 | 0.554 |
| 245  | YNR012W | URK1  | 0         | 0.574 | -0.182 | 0.387 |
| 246  | YDR173C | ARG82 | 0         | 0.558 | -0.182 | 0.552 |
| 247  | YBR159W | IFA38 | 0         | 0.557 | -0.181 | 0.556 |
| 248  | YGL027C | CWH41 | 0         | 0.555 | -0.182 | 0.554 |
| 249  | YDR093W | DNF2  | 0         | 0.552 | -0.182 | 0.500 |
| 250  | YIR029W | DAL2  | 0         | 0.552 | -0.182 | 0.519 |
| 251  | YGR007W | ECT1  | 0         | 0.550 | -0.182 | 0.494 |
| 252  | YJL100W | LSB6  | 0         | 0.549 | -0.182 | 0.412 |
| 253  | YJL200C | ACO2  | 0         | 0.548 | -0.182 | 0.511 |
| 254  | YEL021W | URA3  | 0         | 0.548 | -0.182 | 0.509 |
| 255  | YHR072W | ERG7  | 1         | 0.548 | -0.182 | 0.539 |
| 256  | YGR046W | TAM41 | 1         | 0.546 | -0.182 | 0.544 |
| 257  | YKL087C | CYT2  | 0         | 0.546 | -0.182 | 0.399 |
| 258  | YAL039C | CYC3  | 0         | 0.546 | -0.182 | 0.399 |
| 259  | YML120C | NDI1  | 0         | 0.544 | -0.182 | 0.338 |
| 260  | YDR294C | DPL1  | 0         | 0.543 | -0.182 | 0.540 |
| 261  | YDR321W | ASP1  | 0         | 0.542 | -0.182 | 0.459 |
| 262  | YDR050C | TPI1  | 1         | 0.542 | -0.182 | 0.499 |
| 263  | YGR234W | YHB1  | 0         | 0.540 | -0.182 | 0.560 |
| 264  | YGR175C | ERG1  | 1         | 0.540 | -0.182 | 0.560 |



Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 265  | YDR036C | EHD3  | 0         | 0.540 | -0.182 | 0.560 |
| 266  | YIR028W | DAL4  | 0         | 0.536 | -0.182 | 0.510 |
| 267  | YKL055C | OAR1  | 0         | 0.535 | -0.182 | 0.546 |
| 268  | YBR026C | ETR1  | 0         | 0.535 | -0.182 | 0.546 |
| 269  | YEL006W | YEA6  | 0         | 0.527 | -0.182 | 0.372 |
| 270  | YJL097W | PHS1  | 1         | 0.525 | -0.181 | 0.565 |
| 271  | YJR152W | DAL5  | 0         | 0.525 | -0.182 | 0.422 |
| 272  | YBR235W | VHC1  | 0         | 0.524 | -0.182 | 0.475 |
| 273  | YBR196C | PGI1  | 1         | 0.522 | -0.182 | 0.458 |
| 274  | YKL188C | PXA2  | 0         | 0.514 | -0.182 | 0.471 |
| 275  | YPL147W | PXA1  | 0         | 0.514 | -0.182 | 0.471 |
| 276  | YOL065C | INP54 | 0         | 0.509 | -0.182 | 0.562 |
| 277  | YDR403W | DIT1  | 0         | 0.504 | -0.182 | 0.413 |
| 278  | YLR189C | ATG26 | 0         | 0.501 | -0.182 | 0.405 |
| 279  | YOR316C | COT1  | 0         | 0.500 | -0.182 | 0.367 |
| 280  | YMR243C | ZRC1  | 0         | 0.500 | -0.182 | 0.367 |
| 281  | YDR071C | PAA1  | 0         | 0.498 | -0.182 | 0.415 |
| 282  | YCL030C | HIS4  | 0         | 0.496 | -0.182 | 0.491 |
| 283  | YMR202W | ERG2  | 0         | 0.494 | -0.182 | 0.539 |
| 284  | YBR252W | DUT1  | 1         | 0.493 | -0.182 | 0.385 |
| 285  | YJR024C | MDE1  | 0         | 0.492 | -0.182 | 0.544 |
| 286  | YGL169W | SUA5  | 1         | 0.490 | -0.182 | 0.479 |
| 287  | YBR038W | CHS2  | 1         | 0.487 | -0.182 | 0.210 |
| 288  | YBR023C | CHS3  | 0         | 0.487 | -0.182 | 0.210 |
| 289  | YNL192W | CHS1  | 0         | 0.487 | -0.182 | 0.210 |
| 290  | YDR302W | GPI11 | 1         | 0.485 | -0.182 | 0.399 |
| 291  | YLL031C | GPI13 | 1         | 0.485 | -0.182 | 0.399 |
| 292  | YNL073W | MSK1  | 0         | 0.482 | -0.182 | 0.554 |
| 293  | YHR091C | MSR1  | 0         | 0.482 | -0.182 | 0.554 |
| 294  | YOL143C | RIB4  | 0         | 0.482 | -0.182 | 0.549 |
| 295  | YMR009W | ADI1  | 0         | 0.480 | -0.182 | 0.461 |
| 296  | YEL038W | UTR4  | 0         | 0.480 | -0.182 | 0.461 |
| 297  | YGR157W | CHO2  | 0         | 0.476 | 0.100  | 0.962 |
| 298  | YGR240C | PFK1  | 0         | 0.476 | -0.182 | 0.481 |
| 299  | YMR205C | PFK2  | 0         | 0.476 | -0.182 | 0.481 |
| 300  | YDR147W | EKI1  | 0         | 0.474 | -0.182 | 0.394 |
| 301  | YCL038C | ATG22 | 0         | 0.468 | -0.182 | 0.423 |
| 302  | YHR067W | HTD2  | 0         | 0.461 | -0.182 | 0.551 |
| 303  | YIL155C | GUT2  | 0         | 0.461 | -0.182 | 0.554 |
| 304  | YNL141W | AAH1  | 0         | 0.460 | -0.182 | 0.506 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name    | essential | GCN   | FBA    | LR    |
|------|-----------|---------|-----------|-------|--------|-------|
| 305  | YHR044C   | DOG1    | 0         | 0.457 | -0.182 | 0.409 |
| 306  | YHR043C   | DOG2    | 0         | 0.457 | -0.182 | 0.409 |
| 307  | YER170W   | ADK2    | 0         | 0.457 | -0.182 | 0.550 |
| 308  | YBR022W   | POA1    | 0         | 0.457 | -0.182 | 0.404 |
| 309  | YMR087W   | YMR087W | 0         | 0.457 | -0.182 | 0.404 |
| 310  | YOL058W   | ARG1    | 0         | 0.456 | -0.158 | 0.585 |
| 311  | YDR017C   | KCS1    | 0         | 0.456 | -0.182 | 0.552 |
| 312  | YOR360C   | PDE2    | 0         | 0.455 | -0.182 | 0.422 |
| 313  | YJR078W   | BNA2    | 0         | 0.454 | -0.182 | 0.553 |
| 314  | YIL006W   | YIA6    | 0         | 0.453 | -0.182 | 0.481 |
| 315  | YOR128C   | ADE2    | 0         | 0.452 | -0.182 | 0.544 |
| 316  | YKL094W   | YJU3    | 0         | 0.449 | -0.182 | 0.518 |
| 317  | YGR065C   | VHT1    | 1         | 0.446 | -0.182 | 0.505 |
| 318  | YBL033C   | RIB1    | 0         | 0.446 | -0.182 | 0.435 |
| 319  | YNL169C   | PSD1    | 0         | 0.446 | -0.182 | 0.577 |
| 320  | YDR353W   | TRR1    | 1         | 0.442 | -0.182 | 0.255 |
| 321  | YKL085W   | MDH1    | 0         | 0.440 | -0.182 | 0.520 |
| 322  | YNR056C   | BIO5    | 0         | 0.438 | -0.182 | 0.510 |
| 323  | YCL004W   | PGS1    | 0         | 0.437 | -0.182 | 0.535 |
| 324  | YDL015C   | TSC13   | 1         | 0.436 | -0.181 | 0.506 |
| 325  | YLR410W   | VIP1    | 0         | 0.435 | -0.182 | 0.537 |
| 326  | YOR163W   | DDP1    | 0         | 0.435 | -0.182 | 0.537 |
| 327  | YDR234W   | LYS4    | 0         | 0.433 | -0.052 | 0.840 |
| 328  | YNL277W   | MET2    | 0         | 0.433 | -0.182 | 0.425 |
| 329  | YGR096W   | TPC1    | 0         | 0.431 | -0.182 | 0.510 |
| 330  | YKR031C   | SPO14   | 0         | 0.428 | -0.182 | 0.552 |
| 331  | YIL094C   | LYS12   | 0         | 0.423 | -0.052 | 0.844 |
| 332  | YCR034W   | ELO2    | 0         | 0.420 | -0.181 | 0.484 |
| 333  | YLL012W   | YEH1    | 0         | 0.416 | -0.182 | 0.509 |
| 334  | YDR402C   | DIT2    | 0         | 0.415 | -0.182 | 0.428 |
| 335  | YKL103C   | APE1    | 0         | 0.413 | -0.182 | 0.530 |
| 336  | YMR226C   | YMR226C | 0         | 0.411 | -0.182 | 0.475 |
| 337  | YHR128W   | FUR1    | 1         | 0.410 | -0.182 | 0.405 |
| 338  | YPL087W   | YDC1    | 0         | 0.407 | -0.182 | 0.498 |
| 339  | YBL091C   | MAP2    | 0         | 0.406 | -0.182 | 0.404 |
| 340  | YLR244C   | MAP1    | 0         | 0.406 | -0.182 | 0.404 |
| 341  | Q0080     | ATP8    | 0         | 0.404 | -0.006 | 0.447 |
| 342  | YOL077W-A | ATP19   | 0         | 0.404 | -0.006 | 0.447 |
| 343  | YDR377W   | ATP17   | 0         | 0.404 | -0.006 | 0.447 |
| 344  | Q0130     | OLI1    | 0         | 0.404 | -0.006 | 0.447 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name    | essential | GCN   | FBA    | LR    |
|------|-----------|---------|-----------|-------|--------|-------|
| 345  | YBL099W   | ATP1    | 0         | 0.404 | -0.006 | 0.447 |
| 346  | YML081C-A | ATP18   | 0         | 0.404 | -0.006 | 0.447 |
| 347  | YBR039W   | ATP3    | 0         | 0.404 | -0.006 | 0.447 |
| 348  | YPL271W   | ATP15   | 0         | 0.404 | -0.006 | 0.447 |
| 349  | YDL004W   | ATP16   | 1         | 0.404 | -0.006 | 0.447 |
| 350  | YKL016C   | ATP7    | 0         | 0.404 | -0.006 | 0.447 |
| 351  | YDR298C   | ATP5    | 0         | 0.404 | -0.006 | 0.447 |
| 352  | YPL078C   | ATP4    | 0         | 0.404 | -0.006 | 0.447 |
| 353  | YLR295C   | ATP14   | 0         | 0.404 | -0.006 | 0.447 |
| 354  | YJR121W   | ATP2    | 0         | 0.404 | -0.006 | 0.447 |
| 355  | YPR020W   | ATP20   | 0         | 0.404 | -0.006 | 0.447 |
| 356  | YDR322C-A | TIM11   | 0         | 0.404 | -0.006 | 0.447 |
| 357  | Q0085     | ATP6    | 0         | 0.404 | -0.006 | 0.447 |
| 358  | YKL212W   | SAC1    | 0         | 0.401 | -0.182 | 0.241 |
| 359  | YFR030W   | MET10   | 0         | 0.400 | -0.182 | 0.476 |
| 360  | YJR137C   | MET5    | 0         | 0.400 | -0.182 | 0.476 |
| 361  | YHR018C   | ARG4    | 0         | 0.400 | -0.158 | 0.590 |
| 362  | YLR020C   | YEH2    | 0         | 0.399 | -0.182 | 0.535 |
| 363  | YJR131W   | MNS1    | 0         | 0.399 | -0.182 | 0.393 |
| 364  | YLR057W   | MNL2    | 0         | 0.399 | -0.182 | 0.393 |
| 365  | YGL037C   | PNC1    | 0         | 0.392 | -0.182 | 0.534 |
| 366  | YDR058C   | TGL2    | 0         | 0.391 | -0.182 | 0.562 |
| 367  | YIL010W   | DOT5    | 0         | 0.391 | -0.182 | 0.362 |
| 368  | YDR297W   | SUR2    | 0         | 0.391 | -0.182 | 0.496 |
| 369  | YPR081C   | GRS2    | 0         | 0.389 | -0.182 | 0.383 |
| 370  | YBR121C   | GRS1    | 0         | 0.389 | -0.182 | 0.383 |
| 371  | YGR192C   | TDH3    | 0         | 0.386 | -0.182 | 0.226 |
| 372  | YJL052W   | TDH1    | 0         | 0.386 | -0.182 | 0.226 |
| 373  | YJR009C   | TDH2    | 0         | 0.386 | -0.182 | 0.226 |
| 374  | YPR160W   | GPH1    | 0         | 0.385 | -0.182 | 0.509 |
| 375  | YOR108W   | LEU9    | 0         | 0.382 | -0.182 | 0.320 |
| 376  | YDL219W   | DTD1    | 0         | 0.381 | -0.182 | 0.475 |
| 377  | YDR315C   | IPK1    | 0         | 0.379 | -0.182 | 0.554 |
| 378  | YNL241C   | ZWF1    | 0         | 0.378 | -0.182 | 0.518 |
| 379  | YBR018C   | GAL7    | 0         | 0.374 | -0.182 | 0.402 |
| 380  | YER055C   | HIS1    | 0         | 0.373 | -0.182 | 0.474 |
| 381  | YPR026W   | ATH1    | 0         | 0.370 | -0.182 | 0.399 |
| 382  | YBR001C   | NTH2    | 0         | 0.370 | -0.182 | 0.399 |
| 383  | YMR099C   | YMR099C | 0         | 0.369 | -0.182 | 0.554 |
| 384  | YDR023W   | SES1    | 1         | 0.369 | -0.182 | 0.378 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 385  | YHR011W | DIA4    | 0         | 0.369 | -0.182 | 0.378 |
| 386  | YDR539W | FDC1    | 0         | 0.369 | -0.182 | 0.506 |
| 387  | YHR025W | THR1    | 0         | 0.368 | -0.113 | 0.745 |
| 388  | YNR050C | LYS9    | 0         | 0.362 | -0.052 | 0.658 |
| 389  | YGR260W | TNA1    | 0         | 0.361 | -0.182 | 0.505 |
| 390  | YML008C | ERG6    | 0         | 0.361 | -0.182 | 0.459 |
| 391  | YLL041C | SDH2    | 0         | 0.359 | -0.177 | 0.142 |
| 392  | YKL141W | SDH3    | 1         | 0.359 | -0.177 | 0.142 |
| 393  | YBR042C | CST26   | 0         | 0.359 | -0.182 | 0.500 |
| 394  | YGR208W | SER2    | 0         | 0.358 | -0.180 | 0.523 |
| 395  | YEL029C | BUD16   | 0         | 0.357 | -0.182 | 0.544 |
| 396  | YNR001C | CIT1    | 0         | 0.356 | -0.182 | 0.296 |
| 397  | YLR017W | MEU1    | 0         | 0.355 | -0.182 | 0.509 |
| 398  | YBR036C | CSG2    | 0         | 0.351 | -0.182 | 0.228 |
| 399  | YLR245C | CDD1    | 0         | 0.349 | -0.182 | 0.515 |
| 400  | YKL067W | YNK1    | 0         | 0.349 | 0.100  | 0.806 |
| 401  | YBR176W | ECM31   | 0         | 0.339 | -0.182 | 0.504 |
| 402  | YBR084W | MIS1    | 0         | 0.336 | -0.182 | 0.516 |
| 403  | YBR183W | YPC1    | 0         | 0.335 | -0.182 | 0.499 |
| 404  | YOL011W | PLB3    | 0         | 0.334 | -0.182 | 0.448 |
| 405  | YNR058W | BIO3    | 0         | 0.333 | -0.182 | 0.474 |
| 406  | YCL047C | POF1    | 0         | 0.332 | -0.182 | 0.404 |
| 407  | YDR487C | RIB3    | 1         | 0.327 | -0.182 | 0.464 |
| 408  | YJR019C | TES1    | 0         | 0.326 | -0.182 | 0.540 |
| 409  | YNL202W | SPS19   | 0         | 0.326 | -0.182 | 0.550 |
| 410  | YOR180C | DCI1    | 0         | 0.326 | -0.182 | 0.550 |
| 411  | YPR062W | FCY1    | 0         | 0.324 | -0.182 | 0.505 |
| 412  | YGR194C | XKS1    | 0         | 0.322 | -0.182 | 0.539 |
| 413  | YLR377C | FBP1    | 0         | 0.322 | -0.182 | 0.474 |
| 414  | YFR044C | DUG1    | 0         | 0.322 | -0.182 | 0.401 |
| 415  | YJL196C | ELO1    | 0         | 0.320 | -0.182 | 0.530 |
| 416  | YGL125W | MET13   | 0         | 0.316 | -0.173 | 0.425 |
| 417  | YNL036W | NCE103  | 1         | 0.315 | -0.182 | 0.555 |
| 418  | YEL046C | GLY1    | 0         | 0.315 | -0.182 | 0.397 |
| 419  | YDR009W | GAL3    | 0         | 0.313 | -0.182 | 0.549 |
| 420  | YML035C | AMD1    | 0         | 0.313 | -0.182 | 0.509 |
| 421  | YAR015W | ADE1    | 0         | 0.313 | -0.182 | 0.479 |
| 422  | YBL013W | FMT1    | 0         | 0.311 | -0.182 | 0.514 |
| 423  | YDR178W | SDH4    | 0         | 0.308 | -0.182 | 0.139 |
| 424  | YJL045W | YJL045W | 0         | 0.308 | -0.182 | 0.139 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 425  | YOR155C | ISN1    | 0         | 0.304 | -0.182 | 0.499 |
| 426  | YLR109W | AHP1    | 0         | 0.302 | -0.182 | 0.371 |
| 427  | YPL111W | CAR1    | 0         | 0.301 | -0.182 | 0.464 |
| 428  | YLR372W | ELO3    | 0         | 0.300 | -0.182 | 0.474 |
| 429  | YKL184W | SPE1    | 0         | 0.299 | -0.182 | 0.514 |
| 430  | YNL220W | ADE12   | 0         | 0.295 | -0.182 | 0.340 |
| 431  | YOR209C | NPT1    | 0         | 0.294 | -0.182 | 0.417 |
| 432  | YMR162C | DNF3    | 0         | 0.294 | -0.182 | 0.331 |
| 433  | YBR147W | RTC2    | 0         | 0.294 | -0.182 | 0.248 |
| 434  | YCR036W | RBK1    | 0         | 0.292 | -0.182 | 0.470 |
| 435  | YIL023C | YKE4    | 0         | 0.288 | -0.182 | 0.505 |
| 436  | YKR093W | PTR2    | 0         | 0.286 | -0.182 | 0.303 |
| 437  | YJR133W | XPT1    | 0         | 0.286 | -0.182 | 0.454 |
| 438  | YIL124W | AYR1    | 0         | 0.283 | -0.182 | 0.488 |
| 439  | YIL160C | POT1    | 0         | 0.282 | -0.181 | 0.549 |
| 440  | YLR284C | ECI1    | 0         | 0.282 | -0.182 | 0.540 |
| 441  | YNR027W | BUD17   | 0         | 0.282 | -0.182 | 0.552 |
| 442  | YBR041W | FAT1    | 0         | 0.281 | -0.161 | 0.205 |
| 443  | YGR171C | MSM1    | 0         | 0.280 | -0.182 | 0.549 |
| 444  | YDR400W | URH1    | 0         | 0.277 | -0.182 | 0.455 |
| 445  | YKL127W | PGM1    | 0         | 0.276 | -0.182 | 0.295 |
| 446  | YMR105C | PGM2    | 0         | 0.276 | -0.182 | 0.295 |
| 447  | YER042W | MXR1    | 0         | 0.276 | -0.182 | 0.415 |
| 448  | YKL069W | YKL069W | 0         | 0.276 | -0.182 | 0.415 |
| 449  | YBR263W | SHM1    | 0         | 0.275 | -0.182 | 0.504 |
| 450  | YGL234W | ADE5,7  | 0         | 0.275 | -0.182 | 0.460 |
| 451  | YCL009C | ILV6    | 0         | 0.274 | -0.182 | 0.332 |
| 452  | YLR164W | SHH4    | 0         | 0.274 | -0.182 | 0.142 |
| 453  | YKL148C | SDH1    | 0         | 0.274 | -0.182 | 0.142 |
| 454  | YOR388C | FDH1    | 0         | 0.274 | -0.182 | 0.548 |
| 455  | YGL026C | TRP5    | 0         | 0.273 | -0.182 | 0.363 |
| 456  | YBR034C | HMT1    | 0         | 0.273 | -0.182 | 0.425 |
| 457  | YKL001C | MET14   | 0         | 0.271 | -0.182 | 0.534 |
| 458  | YPL023C | MET12   | 0         | 0.269 | -0.182 | 0.410 |
| 459  | YJL153C | INO1    | 0         | 0.269 | -0.182 | 0.484 |
| 460  | YOR175C | ALE1    | 0         | 0.269 | -0.182 | 0.494 |
| 461  | YDL100C | GET3    | 0         | 0.269 | -0.182 | 0.597 |
| 462  | YPL206C | PGC1    | 0         | 0.268 | -0.182 | 0.550 |
| 463  | YHR100C | GEP4    | 0         | 0.268 | -0.182 | 0.550 |
| 464  | YIL002C | INP51   | 0         | 0.268 | -0.182 | 0.169 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 465  | YJL121C | RPE1  | 0         | 0.265 | -0.182 | 0.546 |
| 466  | YBL064C | PRX1  | 0         | 0.265 | -0.182 | 0.501 |
| 467  | YDL238C | GUD1  | 0         | 0.264 | -0.182 | 0.529 |
| 468  | YDR256C | CTA1  | 0         | 0.263 | -0.182 | 0.583 |
| 469  | YKR043C | SHB17 | 0         | 0.262 | -0.182 | 0.534 |
| 470  | YCR083W | TRX3  | 0         | 0.261 | -0.182 | 0.488 |
| 471  | YJR110W | YMR1  | 0         | 0.260 | -0.182 | 0.245 |
| 472  | YDR158W | HOM2  | 0         | 0.259 | -0.113 | 0.749 |
| 473  | YJL088W | ARG3  | 0         | 0.259 | -0.158 | 0.619 |
| 474  | YMR250W | GAD1  | 0         | 0.259 | -0.182 | 0.293 |
| 475  | YBR180W | DTR1  | 0         | 0.259 | -0.182 | 0.505 |
| 476  | YLR286C | CTS1  | 0         | 0.259 | -0.182 | 0.385 |
| 477  | YMR267W | PPA2  | 0         | 0.258 | -0.179 | 0.578 |
| 478  | YGL205W | POX1  | 0         | 0.258 | -0.181 | 0.544 |
| 479  | YHR046C | INM1  | 0         | 0.256 | -0.182 | 0.394 |
| 480  | YDR287W | INM2  | 0         | 0.256 | -0.182 | 0.394 |
| 481  | YDR226W | ADK1  | 0         | 0.250 | -0.182 | 0.435 |
| 482  | YDR354W | TRP4  | 0         | 0.250 | -0.182 | 0.459 |
| 483  | YJR001W | AVT1  | 0         | 0.249 | -0.182 | 0.299 |
| 484  | YOR136W | IDH2  | 0         | 0.247 | -0.180 | 0.451 |
| 485  | YNL037C | IDH1  | 0         | 0.247 | -0.180 | 0.451 |
| 486  | YJR010W | MET3  | 0         | 0.246 | -0.182 | 0.529 |
| 487  | YAL023C | PMT2  | 0         | 0.244 | -0.182 | 0.151 |
| 488  | YDL095W | PMT1  | 0         | 0.244 | -0.182 | 0.151 |
| 489  | YMR272C | SCS7  | 0         | 0.243 | -0.182 | 0.525 |
| 490  | YKL194C | MST1  | 0         | 0.241 | -0.182 | 0.549 |
| 491  | YLR351C | NIT3  | 0         | 0.240 | -0.182 | 0.435 |
| 492  | YBL042C | FUI1  | 0         | 0.235 | -0.182 | 0.486 |
| 493  | YKR097W | PCK1  | 0         | 0.232 | -0.182 | 0.469 |
| 494  | YAL022C | FUN26 | 0         | 0.230 | -0.182 | 0.466 |
| 495  | YMR165C | PAH1  | 0         | 0.226 | -0.182 | 0.518 |
| 496  | YGR256W | GND2  | 0         | 0.226 | -0.182 | 0.398 |
| 497  | YHR183W | GND1  | 0         | 0.226 | -0.182 | 0.398 |
| 498  | YCR024C | SLM5  | 0         | 0.226 | -0.182 | 0.554 |
| 499  | YOR306C | MCH5  | 0         | 0.225 | -0.182 | 0.524 |
| 500  | YKL215C | OXF1  | 0         | 0.225 | -0.182 | 0.301 |
| 501  | YLR231C | BNA5  | 0         | 0.223 | -0.182 | 0.502 |
| 502  | YBL011W | SCT1  | 0         | 0.223 | -0.182 | 0.359 |
| 503  | YEL041W | YEF1  | 0         | 0.222 | -0.182 | 0.363 |
| 504  | YJR049C | UTR1  | 0         | 0.222 | -0.182 | 0.363 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 505  | YGR155W | CYS4    | 0         | 0.221 | -0.182 | 0.425 |
| 506  | YMR006C | PLB2    | 0         | 0.217 | -0.182 | 0.259 |
| 507  | YIL013C | PDR11   | 0         | 0.215 | -0.182 | 0.112 |
| 508  | YOR011W | AUS1    | 0         | 0.215 | -0.182 | 0.112 |
| 509  | YDR127W | ARO1    | 0         | 0.214 | -0.182 | 0.504 |
| 510  | YOL052C | SPE2    | 0         | 0.213 | -0.182 | 0.474 |
| 511  | YNL268W | LYP1    | 0         | 0.212 | -0.182 | 0.370 |
| 512  | YJR105W | ADO1    | 0         | 0.212 | -0.182 | 0.582 |
| 513  | YPR127W | YPR127W | 0         | 0.210 | -0.182 | 0.583 |
| 514  | YLR354C | TAL1    | 0         | 0.207 | -0.182 | 0.309 |
| 515  | YGR043C | NQM1    | 0         | 0.207 | -0.182 | 0.309 |
| 516  | YPR139C | LOA1    | 0         | 0.206 | -0.182 | 0.363 |
| 517  | YKR080W | MTD1    | 0         | 0.203 | -0.182 | 0.572 |
| 518  | YPL059W | GRX5    | 0         | 0.202 | -0.182 | 0.502 |
| 519  | YER183C | FAU1    | 0         | 0.201 | -0.182 | 0.462 |
| 520  | YDR456W | NHX1    | 0         | 0.201 | -0.182 | 0.379 |
| 521  | YPL274W | SAM3    | 0         | 0.200 | -0.182 | 0.385 |
| 522  | YLR382C | NAM2    | 0         | 0.199 | -0.182 | 0.554 |
| 523  | YPL040C | ISM1    | 0         | 0.199 | -0.182 | 0.554 |
| 524  | YDL022W | GPD1    | 0         | 0.198 | -0.182 | 0.413 |
| 525  | YOR100C | CRC1    | 0         | 0.197 | -0.182 | 0.497 |
| 526  | YGL077C | HNM1    | 0         | 0.197 | -0.182 | 0.323 |
| 527  | YNL117W | MLS1    | 0         | 0.197 | -0.182 | 0.322 |
| 528  | YAL054C | ACS1    | 0         | 0.196 | -0.182 | 0.272 |
| 529  | YML075C | HMG1    | 0         | 0.194 | -0.182 | 0.437 |
| 530  | YLR450W | HMG2    | 0         | 0.194 | -0.182 | 0.437 |
| 531  | YLR237W | THI7    | 0         | 0.194 | -0.182 | 0.268 |
| 532  | YOR071C | NRT1    | 0         | 0.194 | -0.182 | 0.268 |
| 533  | YOR192C | THI72   | 0         | 0.194 | -0.182 | 0.268 |
| 534  | YPL244C | HUT1    | 0         | 0.192 | -0.182 | 0.505 |
| 535  | YPR035W | GLN1    | 1         | 0.191 | -0.182 | 0.285 |
| 536  | YDR408C | ADE8    | 0         | 0.190 | -0.182 | 0.499 |
| 537  | YLR172C | DPH5    | 0         | 0.189 | -0.182 | 0.479 |
| 538  | YHR109W | CTM1    | 0         | 0.189 | -0.182 | 0.479 |
| 539  | YLR285W | NNT1    | 0         | 0.187 | -0.182 | 0.464 |
| 540  | YNL009W | IDP3    | 0         | 0.183 | -0.182 | 0.554 |
| 541  | YLR420W | URA4    | 0         | 0.182 | -0.182 | 0.549 |
| 542  | YIL020C | HIS6    | 0         | 0.182 | -0.182 | 0.549 |
| 543  | YFR025C | HIS2    | 0         | 0.181 | -0.182 | 0.549 |
| 544  | YOR040W | GLO4    | 0         | 0.181 | -0.182 | 0.529 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 545  | YKR009C | FOX2  | 0         | 0.179 | -0.181 | 0.603 |
| 546  | YCR098C | GIT1  | 0         | 0.179 | -0.182 | 0.339 |
| 547  | YGR248W | SOL4  | 0         | 0.179 | -0.182 | 0.422 |
| 548  | YHR163W | SOL3  | 0         | 0.179 | -0.182 | 0.422 |
| 549  | YHR106W | TRR2  | 0         | 0.176 | -0.182 | 0.382 |
| 550  | YMR278W | PRM15 | 0         | 0.176 | -0.182 | 0.259 |
| 551  | YCR028C | FEN2  | 0         | 0.176 | -0.182 | 0.549 |
| 552  | YKL140W | TGL1  | 0         | 0.174 | -0.182 | 0.365 |
| 553  | YOL064C | MET22 | 0         | 0.174 | -0.182 | 0.582 |
| 554  | YIL009W | FAA3  | 0         | 0.173 | -0.182 | 0.334 |
| 555  | YAL012W | CYS3  | 0         | 0.172 | -0.182 | 0.394 |
| 556  | YOR109W | INP53 | 0         | 0.169 | -0.182 | 0.088 |
| 557  | YNL106C | INP52 | 0         | 0.169 | -0.182 | 0.088 |
| 558  | YER081W | SER3  | 0         | 0.169 | -0.182 | 0.299 |
| 559  | YIL074C | SER33 | 0         | 0.169 | -0.182 | 0.299 |
| 560  | YGL248W | PDE1  | 0         | 0.168 | -0.182 | 0.404 |
| 561  | YJR040W | GEF1  | 0         | 0.164 | -0.182 | 0.505 |
| 562  | YGR144W | THI4  | 0         | 0.164 | -0.182 | 0.426 |
| 563  | YOR253W | NAT5  | 0         | 0.164 | -0.182 | 0.339 |
| 564  | YHR013C | ARD1  | 0         | 0.164 | -0.182 | 0.339 |
| 565  | YDL040C | NAT1  | 0         | 0.164 | -0.182 | 0.339 |
| 566  | YMR289W | ABZ2  | 0         | 0.163 | -0.182 | 0.410 |
| 567  | YLR070C | XYL2  | 0         | 0.162 | -0.182 | 0.539 |
| 568  | YEL031W | SPF1  | 0         | 0.162 | -0.182 | 0.510 |
| 569  | YOR311C | DGK1  | 0         | 0.162 | -0.182 | 0.512 |
| 570  | YML056C | IMD4  | 0         | 0.162 | -0.182 | 0.276 |
| 571  | YHR216W | IMD2  | 0         | 0.162 | -0.182 | 0.276 |
| 572  | YLR432W | IMD3  | 0         | 0.162 | -0.182 | 0.276 |
| 573  | YDR007W | TRP1  | 0         | 0.162 | -0.182 | 0.549 |
| 574  | YLR359W | ADE13 | 1         | 0.162 | -0.182 | 0.510 |
| 575  | YGR204W | ADE3  | 0         | 0.161 | -0.182 | 0.465 |
| 576  | YGL196W | DSD1  | 0         | 0.159 | -0.182 | 0.415 |
| 577  | YDR399W | HPT1  | 0         | 0.159 | -0.182 | 0.383 |
| 578  | YDR428C | BNA7  | 0         | 0.156 | -0.182 | 0.528 |
| 579  | YCR068W | ATG15 | 0         | 0.154 | -0.182 | 0.360 |
| 580  | YLR138W | NHA1  | 0         | 0.154 | -0.182 | 0.390 |
| 581  | YBR006W | UGA2  | 0         | 0.151 | -0.182 | 0.499 |
| 582  | YLR180W | SAM1  | 0         | 0.151 | -0.182 | 0.311 |
| 583  | YDR502C | SAM2  | 0         | 0.151 | -0.182 | 0.311 |
| 584  | YPL148C | PPT2  | 0         | 0.150 | -0.182 | 0.597 |



Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 585  | YNL316C | PHA2    | 0         | 0.148 | -0.182 | 0.534 |
| 586  | YBR132C | AGP2    | 0         | 0.147 | -0.182 | 0.444 |
| 587  | YIR032C | DAL3    | 0         | 0.145 | -0.182 | 0.500 |
| 588  | YJR051W | OSM1    | 0         | 0.143 | -0.182 | 0.489 |
| 589  | YKL174C | TPO5    | 0         | 0.143 | -0.182 | 0.461 |
| 590  | YJL134W | LCB3    | 0         | 0.143 | -0.182 | 0.351 |
| 591  | YLR260W | LCB5    | 0         | 0.143 | -0.182 | 0.351 |
| 592  | YKR053C | YSR3    | 0         | 0.143 | -0.182 | 0.351 |
| 593  | YOR171C | LCB4    | 0         | 0.143 | -0.182 | 0.351 |
| 594  | YDR497C | ITR1    | 0         | 0.141 | -0.182 | 0.399 |
| 595  | YOL103W | ITR2    | 0         | 0.141 | -0.182 | 0.399 |
| 596  | YBR166C | TYR1    | 0         | 0.141 | -0.182 | 0.524 |
| 597  | YMR008C | PLB1    | 0         | 0.137 | -0.182 | 0.463 |
| 598  | YNL270C | ALP1    | 0         | 0.137 | -0.182 | 0.252 |
| 599  | YEL063C | CAN1    | 0         | 0.137 | -0.182 | 0.252 |
| 600  | YLR303W | MET17   | 0         | 0.136 | -0.182 | 0.307 |
| 601  | YFR049W | YMR31   | 0         | 0.134 | -0.182 | 0.227 |
| 602  | YLR174W | IDP2    | 0         | 0.134 | -0.182 | 0.463 |
| 603  | YGR061C | ADE6    | 0         | 0.133 | -0.182 | 0.277 |
| 604  | YKL029C | MAE1    | 0         | 0.133 | -0.179 | 0.469 |
| 605  | YER091C | MET6    | 0         | 0.133 | -0.173 | 0.421 |
| 606  | YOR334W | MRS2    | 0         | 0.132 | -0.182 | 0.358 |
| 607  | YPL060W | MFM1    | 0         | 0.132 | -0.182 | 0.358 |
| 608  | YHL016C | DUR3    | 0         | 0.132 | -0.182 | 0.429 |
| 609  | YOL092W | YPQ1    | 0         | 0.131 | -0.182 | 0.277 |
| 610  | YDR352W | YPQ2    | 0         | 0.131 | -0.182 | 0.277 |
| 611  | YGL224C | SDT1    | 0         | 0.128 | -0.182 | 0.342 |
| 612  | YHL032C | GUT1    | 0         | 0.127 | -0.182 | 0.489 |
| 613  | YDL024C | DIA3    | 0         | 0.127 | -0.182 | 0.393 |
| 614  | YOR202W | HIS3    | 0         | 0.127 | -0.182 | 0.549 |
| 615  | YER069W | ARG5,6  | 0         | 0.126 | -0.158 | 0.648 |
| 616  | YER086W | ILV1    | 0         | 0.126 | -0.182 | 0.529 |
| 617  | YER015W | FAA2    | 0         | 0.125 | -0.182 | 0.353 |
| 618  | YBR208C | DUR1,2  | 0         | 0.125 | -0.182 | 0.485 |
| 619  | YML042W | CAT2    | 0         | 0.124 | -0.182 | 0.393 |
| 620  | YKL216W | URA1    | 0         | 0.124 | -0.182 | 0.474 |
| 621  | YHR008C | SOD2    | 0         | 0.124 | -0.182 | 0.529 |
| 622  | YGR012W | YGR012W | 0         | 0.124 | -0.182 | 0.534 |
| 623  | YGL148W | ARO2    | 0         | 0.123 | -0.182 | 0.534 |
| 624  | YNL274C | GOR1    | 0         | 0.121 | -0.182 | 0.529 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 625  | YMR210W | YMR210W | 0         | 0.121 | -0.182 | 0.489 |
| 626  | YBR291C | CTP1    | 0         | 0.121 | -0.182 | 0.366 |
| 627  | YDL142C | CRD1    | 0         | 0.120 | -0.182 | 0.560 |
| 628  | YOR161C | PNS1    | 0         | 0.119 | -0.182 | 0.326 |
| 629  | YDR270W | CCC2    | 0         | 0.118 | -0.182 | 0.510 |
| 630  | YKL218C | SRY1    | 0         | 0.118 | -0.182 | 0.524 |
| 631  | YOR323C | PRO2    | 0         | 0.118 | -0.182 | 0.592 |
| 632  | YJR143C | PMT4    | 0         | 0.117 | -0.182 | 0.161 |
| 633  | YDL093W | PMT5    | 0         | 0.117 | -0.182 | 0.161 |
| 634  | YOR321W | PMT3    | 0         | 0.117 | -0.182 | 0.161 |
| 635  | YLR209C | PNP1    | 0         | 0.115 | -0.182 | 0.306 |
| 636  | YCR053W | THR4    | 0         | 0.114 | -0.113 | 0.596 |
| 637  | YCR005C | CIT2    | 0         | 0.114 | -0.182 | 0.577 |
| 638  | YKL192C | ACP1    | 0         | 0.114 | -0.182 | 0.422 |
| 639  | YMR246W | FAA4    | 0         | 0.113 | -0.182 | 0.104 |
| 640  | YPL104W | MSD1    | 0         | 0.110 | -0.182 | 0.549 |
| 641  | YMR207C | HFA1    | 0         | 0.110 | -0.182 | 0.449 |
| 642  | YKL150W | MCR1    | 0         | 0.109 | -0.182 | 0.280 |
| 643  | YOL126C | MDH2    | 0         | 0.108 | -0.182 | 0.489 |
| 644  | YBR213W | MET8    | 0         | 0.108 | -0.182 | 0.495 |
| 645  | YJL129C | TRK1    | 0         | 0.105 | -0.182 | 0.404 |
| 646  | YPL092W | SSU1    | 0         | 0.103 | -0.182 | 0.485 |
| 647  | YMR145C | NDE1    | 0         | 0.101 | -0.182 | 0.256 |
| 648  | YDL085W | NDE2    | 0         | 0.101 | -0.182 | 0.256 |
| 649  | YGL006W | PMC1    | 0         | 0.099 | -0.182 | 0.505 |
| 650  | YGL167C | PMR1    | 0         | 0.099 | -0.182 | 0.505 |
| 651  | YFR015C | GSY1    | 0         | 0.099 | -0.182 | 0.119 |
| 652  | YLR258W | GSY2    | 0         | 0.099 | -0.182 | 0.119 |
| 653  | YKR058W | GLG1    | 0         | 0.099 | -0.182 | 0.119 |
| 654  | YJL137C | GLG2    | 0         | 0.099 | -0.182 | 0.119 |
| 655  | YBR149W | ARA1    | 0         | 0.097 | -0.182 | 0.539 |
| 656  | YMR041C | ARA2    | 0         | 0.097 | -0.182 | 0.539 |
| 657  | YBR293W | VBA2    | 0         | 0.097 | -0.182 | 0.227 |
| 658  | YMR241W | YHM2    | 0         | 0.095 | -0.182 | 0.306 |
| 659  | YOR348C | PUT4    | 0         | 0.095 | -0.182 | 0.264 |
| 660  | YGL255W | ZRT1    | 0         | 0.094 | -0.182 | 0.404 |
| 661  | YLR130C | ZRT2    | 0         | 0.094 | -0.182 | 0.404 |
| 662  | YJL126W | NIT2    | 0         | 0.094 | -0.182 | 0.307 |
| 663  | YOL151W | GRE2    | 0         | 0.093 | -0.182 | 0.498 |
| 664  | YOR317W | FAA1    | 0         | 0.091 | -0.182 | 0.169 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name   | essential | GCN   | FBA    | LR    |
|------|---------|--------|-----------|-------|--------|-------|
| 665  | YJR025C | BNA1   | 0         | 0.091 | -0.182 | 0.597 |
| 666  | YER065C | ICL1   | 0         | 0.089 | -0.182 | 0.459 |
| 667  | YBL080C | PET112 | 0         | 0.089 | -0.182 | 0.514 |
| 668  | YOL033W | MSE1   | 0         | 0.089 | -0.182 | 0.514 |
| 669  | YBL098W | BNA4   | 0         | 0.089 | -0.182 | 0.587 |
| 670  | YPR128C | ANT1   | 0         | 0.087 | -0.182 | 0.538 |
| 671  | YDL066W | IDP1   | 0         | 0.086 | -0.182 | 0.497 |
| 672  | YPR002W | PDH1   | 0         | 0.085 | -0.182 | 0.592 |
| 673  | YNL325C | FIG4   | 0         | 0.085 | -0.182 | 0.445 |
| 674  | YLR386W | VAC14  | 0         | 0.085 | -0.182 | 0.445 |
| 675  | YNL054W | VAC7   | 0         | 0.085 | -0.182 | 0.445 |
| 676  | YFR021W | ATG18  | 0         | 0.085 | -0.182 | 0.445 |
| 677  | YNR033W | ABZ1   | 0         | 0.083 | -0.182 | 0.273 |
| 678  | YNL003C | PET8   | 0         | 0.081 | -0.182 | 0.411 |
| 679  | YER119C | AVT6   | 0         | 0.081 | -0.182 | 0.247 |
| 680  | YJR130C | STR2   | 0         | 0.079 | -0.182 | 0.349 |
| 681  | YLR348C | DIC1   | 0         | 0.079 | -0.182 | 0.360 |
| 682  | YDR284C | DPP1   | 0         | 0.078 | -0.182 | 0.581 |
| 683  | YDL078C | MDH3   | 0         | 0.075 | -0.181 | 0.590 |
| 684  | YML059C | NTE1   | 0         | 0.074 | -0.182 | 0.542 |
| 685  | YLR328W | NMA1   | 0         | 0.074 | -0.182 | 0.345 |
| 686  | YPR060C | ARO7   | 0         | 0.073 | -0.182 | 0.529 |
| 687  | YFR047C | BNA6   | 0         | 0.071 | -0.182 | 0.448 |
| 688  | YEL066W | HPA3   | 0         | 0.071 | -0.182 | 0.308 |
| 689  | YPR193C | HPA2   | 0         | 0.071 | -0.182 | 0.308 |
| 690  | YGR143W | SKN1   | 0         | 0.071 | -0.182 | 0.282 |
| 691  | YPR159W | KRE6   | 0         | 0.071 | -0.182 | 0.282 |
| 692  | YCL064C | CHA1   | 0         | 0.070 | -0.182 | 0.306 |
| 693  | YOR120W | GCY1   | 0         | 0.066 | -0.182 | 0.494 |
| 694  | YJR095W | SFC1   | 0         | 0.065 | -0.182 | 0.392 |
| 695  | YKR052C | MRS4   | 0         | 0.064 | -0.182 | 0.348 |
| 696  | YJL133W | MRS3   | 0         | 0.064 | -0.182 | 0.348 |
| 697  | YDR452W | PPN1   | 0         | 0.064 | -0.182 | 0.095 |
| 698  | YJR073C | OPI3   | 0         | 0.063 | 0.100  | 0.963 |
| 699  | YMR062C | ARG7   | 0         | 0.060 | -0.158 | 0.483 |
| 700  | YIL107C | PFK26  | 0         | 0.060 | -0.182 | 0.339 |
| 701  | YOL136C | PFK27  | 0         | 0.060 | -0.182 | 0.339 |
| 702  | YJL155C | FBP26  | 0         | 0.060 | -0.182 | 0.339 |
| 703  | YMR261C | TPS3   | 0         | 0.060 | -0.182 | 0.237 |
| 704  | YML100W | TSL1   | 0         | 0.060 | -0.182 | 0.237 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 705  | YGL186C | TPN1  | 0         | 0.059 | -0.175 | 0.508 |
| 706  | YJL012C | VTC4  | 0         | 0.059 | -0.182 | 0.505 |
| 707  | YNL065W | AQR1  | 0         | 0.059 | -0.182 | 0.218 |
| 708  | YBR020W | GAL1  | 0         | 0.058 | -0.182 | 0.519 |
| 709  | YDL215C | GDH2  | 0         | 0.058 | -0.182 | 0.265 |
| 710  | YPL258C | THI21 | 0         | 0.057 | -0.182 | 0.267 |
| 711  | YOL055C | THI20 | 0         | 0.057 | -0.182 | 0.267 |
| 712  | YPR121W | THI22 | 0         | 0.057 | -0.182 | 0.267 |
| 713  | YGR088W | CTT1  | 0         | 0.056 | -0.182 | 0.494 |
| 714  | YJR104C | SOD1  | 0         | 0.056 | -0.182 | 0.494 |
| 715  | YGR032W | GSC2  | 0         | 0.055 | -0.182 | 0.172 |
| 716  | YMR306W | FKS3  | 0         | 0.055 | -0.182 | 0.172 |
| 717  | YLR342W | FKS1  | 0         | 0.055 | -0.182 | 0.172 |
| 718  | YOL059W | GPD2  | 0         | 0.054 | -0.182 | 0.480 |
| 719  | YML106W | URA5  | 0         | 0.053 | -0.182 | 0.330 |
| 720  | YMR271C | URA10 | 0         | 0.053 | -0.182 | 0.330 |
| 721  | YER024W | YAT2  | 0         | 0.052 | -0.182 | 0.300 |
| 722  | YAR035W | YAT1  | 0         | 0.052 | -0.182 | 0.300 |
| 723  | YBR294W | SUL1  | 0         | 0.051 | -0.182 | 0.389 |
| 724  | YLR092W | SUL2  | 0         | 0.051 | -0.182 | 0.389 |
| 725  | YBL015W | ACH1  | 0         | 0.050 | -0.182 | 0.359 |
| 726  | YAL060W | BDH1  | 0         | 0.048 | -0.182 | 0.532 |
| 727  | YLL057C | JLP1  | 0         | 0.048 | -0.182 | 0.354 |
| 728  | YLL028W | TPO1  | 0         | 0.048 | -0.182 | 0.166 |
| 729  | YOR273C | TPO4  | 0         | 0.048 | -0.182 | 0.166 |
| 730  | YBR035C | PDX3  | 0         | 0.047 | -0.182 | 0.458 |
| 731  | YOL140W | ARG8  | 0         | 0.047 | -0.158 | 0.541 |
| 732  | YPR167C | MET16 | 0         | 0.047 | -0.182 | 0.309 |
| 733  | YHR123W | EPT1  | 0         | 0.047 | -0.182 | 0.450 |
| 734  | YNL130C | CPT1  | 0         | 0.046 | -0.182 | 0.320 |
| 735  | YML086C | ALO1  | 0         | 0.044 | -0.182 | 0.484 |
| 736  | YFL030W | AGX1  | 0         | 0.043 | -0.182 | 0.305 |
| 737  | YBR248C | HIS7  | 0         | 0.043 | -0.182 | 0.265 |
| 738  | YGR010W | NMA2  | 0         | 0.042 | -0.182 | 0.367 |
| 739  | YLL051C | FRE6  | 0         | 0.042 | -0.182 | 0.455 |
| 740  | YPR006C | ICL2  | 0         | 0.041 | -0.182 | 0.449 |
| 741  | YBR021W | FUR4  | 0         | 0.040 | -0.180 | 0.528 |
| 742  | YGR110W | CLD1  | 0         | 0.038 | -0.182 | 0.589 |
| 743  | YGL062W | PYC1  | 0         | 0.037 | -0.182 | 0.265 |
| 744  | YBR218C | PYC2  | 0         | 0.037 | -0.182 | 0.265 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 745  | YLR299W | ECM38   | 0         | 0.037 | -0.182 | 0.396 |
| 746  | YNL101W | AVT4    | 0         | 0.037 | -0.182 | 0.178 |
| 747  | YKL146W | AVT3    | 0         | 0.037 | -0.182 | 0.178 |
| 748  | YOR130C | ORT1    | 0         | 0.036 | -0.158 | 0.591 |
| 749  | YKR076W | ECM4    | 0         | 0.034 | -0.182 | 0.454 |
| 750  | YFR019W | FAB1    | 0         | 0.034 | -0.182 | 0.455 |
| 751  | YGR289C | MAL11   | 0         | 0.034 | -0.182 | 0.439 |
| 752  | YGR019W | UGA1    | 0         | 0.034 | -0.182 | 0.249 |
| 753  | YMR020W | FMS1    | 0         | 0.033 | -0.182 | 0.422 |
| 754  | YIL125W | KGD1    | 0         | 0.033 | -0.179 | 0.124 |
| 755  | YDR148C | KGD2    | 0         | 0.033 | -0.179 | 0.124 |
| 756  | YPL188W | POS5    | 0         | 0.033 | -0.182 | 0.283 |
| 757  | YDR384C | ATO3    | 0         | 0.033 | -0.182 | 0.187 |
| 758  | YGR121C | MEP1    | 0         | 0.033 | -0.182 | 0.187 |
| 759  | YPR138C | MEP3    | 0         | 0.033 | -0.182 | 0.187 |
| 760  | YNL142W | MEP2    | 0         | 0.033 | -0.182 | 0.187 |
| 761  | YJL068C | YJL068C | 0         | 0.032 | -0.182 | 0.401 |
| 762  | YDL052C | SLC1    | 0         | 0.032 | -0.182 | 0.279 |
| 763  | YML022W | APT1    | 0         | 0.031 | -0.182 | 0.313 |
| 764  | YDR441C | APT2    | 0         | 0.031 | -0.182 | 0.313 |
| 765  | YER163C | GCG1    | 0         | 0.031 | -0.182 | 0.440 |
| 766  | YJL212C | OPT1    | 0         | 0.030 | -0.182 | 0.412 |
| 767  | YLR089C | ALT1    | 0         | 0.029 | -0.182 | 0.372 |
| 768  | YDL171C | GLT1    | 0         | 0.029 | -0.182 | 0.246 |
| 769  | YLR058C | SHM2    | 0         | 0.028 | -0.182 | 0.400 |
| 770  | YGR177C | ATF2    | 0         | 0.028 | -0.182 | 0.102 |
| 771  | YOR377W | ATF1    | 0         | 0.028 | -0.182 | 0.102 |
| 772  | YDR035W | ARO3    | 0         | 0.027 | -0.182 | 0.295 |
| 773  | YBR249C | ARO4    | 0         | 0.027 | -0.182 | 0.295 |
| 774  | YDR481C | PHO8    | 0         | 0.027 | -0.182 | 0.411 |
| 775  | YOR245C | DGA1    | 0         | 0.026 | -0.182 | 0.248 |
| 776  | YDR135C | YCF1    | 0         | 0.025 | -0.182 | 0.405 |
| 777  | YLR043C | TRX1    | 0         | 0.025 | -0.182 | 0.118 |
| 778  | YIL116W | HIS5    | 0         | 0.024 | -0.182 | 0.257 |
| 779  | YPR058W | YMC1    | 0         | 0.024 | -0.182 | 0.454 |
| 780  | YIL053W | GPP1    | 0         | 0.024 | -0.182 | 0.342 |
| 781  | YER062C | GPP2    | 0         | 0.024 | -0.182 | 0.342 |
| 782  | YLR028C | ADE16   | 0         | 0.023 | -0.182 | 0.311 |
| 783  | YMR120C | ADE17   | 0         | 0.023 | -0.182 | 0.311 |
| 784  | YAL035W | FUN12   | 0         | 0.023 | -0.182 | 0.211 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 785  | YLL001W | DNM1  | 0         | 0.023 | -0.182 | 0.211 |
| 786  | YCR010C | ADY2  | 0         | 0.023 | -0.182 | 0.455 |
| 787  | YBR069C | TAT1  | 0         | 0.021 | -0.182 | 0.132 |
| 788  | YPR140W | TAZ1  | 0         | 0.021 | -0.182 | 0.750 |
| 789  | YBR117C | TKL2  | 0         | 0.020 | -0.182 | 0.237 |
| 790  | YPR074C | TKL1  | 0         | 0.020 | -0.182 | 0.237 |
| 791  | YPL262W | FUM1  | 0         | 0.020 | -0.176 | 0.432 |
| 792  | YLR307W | CDA1  | 0         | 0.020 | -0.182 | 0.296 |
| 793  | YLR308W | CDA2  | 0         | 0.020 | -0.182 | 0.296 |
| 794  | YDR453C | TSA2  | 0         | 0.020 | -0.182 | 0.292 |
| 795  | YLR304C | ACO1  | 0         | 0.020 | -0.182 | 0.489 |
| 796  | YDR038C | ENA5  | 0         | 0.019 | -0.182 | 0.235 |
| 797  | YDR039C | ENA2  | 0         | 0.019 | -0.182 | 0.235 |
| 798  | YDR040C | ENA1  | 0         | 0.019 | -0.182 | 0.235 |
| 799  | YOR285W | RDL1  | 0         | 0.019 | -0.182 | 0.280 |
| 800  | YOR286W | RDL2  | 0         | 0.019 | -0.182 | 0.280 |
| 801  | YBR092C | PHO3  | 0         | 0.019 | -0.182 | 0.395 |
| 802  | YEL047C | FRD1  | 0         | 0.018 | -0.182 | 0.259 |
| 803  | YBR128C | ATG14 | 0         | 0.018 | -0.182 | 0.169 |
| 804  | YML054C | CYB2  | 0         | 0.017 | -0.182 | 0.139 |
| 805  | YMR313C | TGL3  | 0         | 0.017 | -0.182 | 0.305 |
| 806  | YPL120W | VPS30 | 0         | 0.017 | -0.182 | 0.141 |
| 807  | YLR438W | CAR2  | 0         | 0.016 | -0.182 | 0.246 |
| 808  | YJR069C | HAM1  | 0         | 0.016 | -0.182 | 0.361 |
| 809  | YDR272W | GLO2  | 0         | 0.015 | -0.182 | 0.405 |
| 810  | YLR360W | VPS38 | 0         | 0.015 | -0.182 | 0.179 |
| 811  | YPR001W | CIT3  | 0         | 0.015 | -0.182 | 0.295 |
| 812  | YNR008W | LRO1  | 0         | 0.013 | -0.182 | 0.649 |
| 813  | YGL080W | MPC1  | 0         | 0.013 | -0.182 | 0.198 |
| 814  | YKR089C | TGL4  | 0         | 0.013 | -0.182 | 0.289 |
| 815  | YKL217W | JEN1  | 0         | 0.013 | -0.182 | 0.174 |
| 816  | YEL042W | GDA1  | 0         | 0.013 | -0.182 | 0.377 |
| 817  | YHR162W | MPC2  | 0         | 0.012 | -0.182 | 0.202 |
| 818  | YGR243W | MPC3  | 0         | 0.012 | -0.182 | 0.202 |
| 819  | YKR066C | CCP1  | 0         | 0.012 | -0.182 | 0.230 |
| 820  | YKL132C | RMA1  | 0         | 0.012 | -0.182 | 0.173 |
| 821  | YOR241W | MET7  | 0         | 0.012 | -0.182 | 0.173 |
| 822  | YIR031C | DAL7  | 0         | 0.012 | -0.182 | 0.260 |
| 823  | YER178W | PDA1  | 0         | 0.012 | -0.176 | 0.345 |
| 824  | YNL071W | LAT1  | 0         | 0.012 | -0.176 | 0.345 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name  | essential | GCN   | FBA    | LR    |
|------|-----------|-------|-----------|-------|--------|-------|
| 825  | YGR193C   | PDX1  | 0         | 0.012 | -0.176 | 0.345 |
| 826  | YBR221C   | PDB1  | 0         | 0.012 | -0.176 | 0.345 |
| 827  | YDR300C   | PRO1  | 0         | 0.011 | -0.182 | 0.339 |
| 828  | YMR058W   | FET3  | 0         | 0.011 | -0.182 | 0.357 |
| 829  | YMR319C   | FET4  | 0         | 0.011 | -0.182 | 0.357 |
| 830  | YGL184C   | STR3  | 0         | 0.011 | -0.182 | 0.245 |
| 831  | YML004C   | GLO1  | 0         | 0.010 | -0.182 | 0.425 |
| 832  | YCL050C   | APA1  | 0         | 0.010 | -0.182 | 0.311 |
| 833  | YBR097W   | VPS15 | 0         | 0.009 | -0.182 | 0.118 |
| 834  | YLR240W   | VPS34 | 0         | 0.009 | -0.182 | 0.118 |
| 835  | YPL273W   | SAM4  | 0         | 0.009 | -0.182 | 0.248 |
| 836  | YLL062C   | MHT1  | 0         | 0.009 | -0.182 | 0.248 |
| 837  | YMR300C   | ADE4  | 0         | 0.009 | -0.182 | 0.221 |
| 838  | YPL091W   | GLR1  | 0         | 0.009 | -0.182 | 0.340 |
| 839  | YJR159W   | SOR1  | 0         | 0.007 | -0.182 | 0.438 |
| 840  | YER060W-A | FCY22 | 0         | 0.007 | -0.182 | 0.211 |
| 841  | YER060W   | FCY21 | 0         | 0.007 | -0.182 | 0.211 |
| 842  | YER056C   | FCY2  | 0         | 0.007 | -0.182 | 0.211 |
| 843  | YPR156C   | TPO3  | 0         | 0.007 | -0.182 | 0.079 |
| 844  | YGR138C   | TPO2  | 0         | 0.007 | -0.182 | 0.079 |
| 845  | YHR174W   | ENO2  | 0         | 0.007 | -0.182 | 0.095 |
| 846  | YGR254W   | ENO1  | 0         | 0.007 | -0.182 | 0.095 |
| 847  | YMR323W   | ERR3  | 0         | 0.007 | -0.182 | 0.095 |
| 848  | YOR393W   | ERR1  | 0         | 0.007 | -0.182 | 0.095 |
| 849  | YPL281C   | ERR2  | 0         | 0.007 | -0.182 | 0.095 |
| 850  | YLL043W   | FPS1  | 0         | 0.007 | -0.182 | 0.308 |
| 851  | YGR255C   | COQ6  | 0         | 0.007 | -0.182 | 0.331 |
| 852  | YDR204W   | COQ4  | 0         | 0.007 | -0.182 | 0.331 |
| 853  | YLR201C   | COQ9  | 0         | 0.007 | -0.182 | 0.331 |
| 854  | YOL096C   | COQ3  | 0         | 0.007 | -0.182 | 0.331 |
| 855  | YOR125C   | CAT5  | 0         | 0.007 | -0.182 | 0.331 |
| 856  | YML110C   | COQ5  | 0         | 0.007 | -0.182 | 0.331 |
| 857  | YKL120W   | OAC1  | 0         | 0.006 | -0.182 | 0.415 |
| 858  | YHR201C   | PPX1  | 0         | 0.006 | -0.182 | 0.068 |
| 859  | YOR251C   | TUM1  | 0         | 0.006 | -0.182 | 0.293 |
| 860  | YBR019C   | GAL10 | 0         | 0.005 | -0.182 | 0.336 |
| 861  | YMR303C   | ADH2  | 0         | 0.005 | -0.182 | 0.484 |
| 862  | YOR142W   | LSC1  | 0         | 0.005 | -0.179 | 0.330 |
| 863  | YGR244C   | LSC2  | 0         | 0.005 | -0.179 | 0.330 |
| 864  | YAL062W   | GDH3  | 0         | 0.004 | -0.182 | 0.190 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name    | essential | GCN   | FBA    | LR    |
|------|-----------|---------|-----------|-------|--------|-------|
| 865  | YOR375C   | GDH1    | 0         | 0.004 | -0.182 | 0.190 |
| 866  | YDL131W   | LYS21   | 0         | 0.004 | -0.182 | 0.224 |
| 867  | YDL182W   | LYS20   | 0         | 0.004 | -0.182 | 0.224 |
| 868  | YDR380W   | ARO10   | 0         | 0.004 | -0.182 | 0.553 |
| 869  | YML070W   | DAK1    | 0         | 0.004 | -0.182 | 0.215 |
| 870  | YFL053W   | DAK2    | 0         | 0.004 | -0.182 | 0.215 |
| 871  | YJL071W   | ARG2    | 0         | 0.004 | -0.182 | 0.279 |
| 872  | YPR184W   | GDB1    | 0         | 0.004 | -0.182 | 0.368 |
| 873  | YGR124W   | ASN2    | 0         | 0.003 | -0.182 | 0.187 |
| 874  | YPR145W   | ASN1    | 0         | 0.003 | -0.182 | 0.187 |
| 875  | YBR296C   | PHO89   | 0         | 0.003 | -0.182 | 0.118 |
| 876  | YJR103W   | URA8    | 0         | 0.003 | -0.182 | 0.154 |
| 877  | YBL039C   | URA7    | 0         | 0.003 | -0.182 | 0.154 |
| 878  | YLL015W   | BPT1    | 0         | 0.003 | -0.182 | 0.280 |
| 879  | YKL181W   | PRS1    | 0         | 0.003 | -0.182 | 0.076 |
| 880  | YDL178W   | DLD2    | 0         | 0.003 | -0.182 | 0.148 |
| 881  | YDR001C   | NTH1    | 0         | 0.003 | -0.182 | 0.368 |
| 882  | YOL061W   | PRS5    | 0         | 0.003 | -0.182 | 0.078 |
| 883  | YER099C   | PRS2    | 0         | 0.003 | -0.182 | 0.078 |
| 884  | YBL068W   | PRS4    | 0         | 0.003 | -0.182 | 0.078 |
| 885  | YBL045C   | COR1    | 0         | 0.003 | 0.044  | 0.358 |
| 886  | YPR191W   | QCR2    | 0         | 0.003 | 0.044  | 0.358 |
| 887  | Q0105     | COB     | 0         | 0.003 | 0.044  | 0.358 |
| 888  | YGR183C   | QCR9    | 0         | 0.003 | 0.044  | 0.358 |
| 889  | YEL024W   | RIP1    | 0         | 0.003 | 0.044  | 0.358 |
| 890  | YFR033C   | QCR6    | 0         | 0.003 | 0.044  | 0.358 |
| 891  | YDR529C   | QCR7    | 0         | 0.003 | 0.044  | 0.358 |
| 892  | YHR001W-A | QCR10   | 0         | 0.003 | 0.044  | 0.358 |
| 893  | YJL166W   | QCR8    | 0         | 0.003 | 0.044  | 0.358 |
| 894  | YOR065W   | CYT1    | 0         | 0.003 | 0.044  | 0.358 |
| 895  | YHL011C   | PRS3    | 0         | 0.003 | -0.182 | 0.080 |
| 896  | YOR081C   | TGL5    | 0         | 0.003 | -0.182 | 0.111 |
| 897  | YER152C   | YER152C | 0         | 0.003 | -0.182 | 0.116 |
| 898  | YER175C   | TMT1    | 0         | 0.003 | -0.182 | 0.350 |
| 899  | YDL080C   | THI3    | 0         | 0.002 | -0.182 | 0.493 |
| 900  | YJL130C   | URA2    | 0         | 0.002 | -0.182 | 0.239 |
| 901  | YDR538W   | PAD1    | 0         | 0.002 | -0.182 | 0.384 |
| 902  | YNL104C   | LEU4    | 0         | 0.002 | -0.182 | 0.321 |
| 903  | YJR148W   | BAT2    | 0         | 0.002 | -0.038 | 0.376 |
| 904  | YOR184W   | SER1    | 0         | 0.002 | -0.180 | 0.300 |



Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name    | essential | GCN   | FBA    | LR    |
|------|---------|---------|-----------|-------|--------|-------|
| 905  | YPL095C | EEB1    | 0         | 0.002 | -0.182 | 0.365 |
| 906  | YBR177C | EHT1    | 0         | 0.002 | -0.182 | 0.365 |
| 907  | YDR019C | GCV1    | 0         | 0.002 | -0.182 | 0.295 |
| 908  | YAL044C | GCV3    | 0         | 0.002 | -0.182 | 0.295 |
| 909  | YMR189W | GCV2    | 0         | 0.002 | -0.182 | 0.295 |
| 910  | YPR069C | SPE3    | 0         | 0.002 | -0.182 | 0.365 |
| 911  | YIL162W | SUC2    | 0         | 0.002 | -0.182 | 0.425 |
| 912  | YJR109C | CPA2    | 0         | 0.002 | -0.182 | 0.173 |
| 913  | YOR303W | CPA1    | 0         | 0.002 | -0.182 | 0.173 |
| 914  | YNR019W | ARE2    | 0         | 0.002 | -0.182 | 0.110 |
| 915  | YPR021C | AGC1    | 0         | 0.002 | -0.181 | 0.231 |
| 916  | YDL067C | COX9    | 0         | 0.002 | 0.045  | 0.051 |
| 917  | Q0045   | COX1    | 0         | 0.002 | 0.045  | 0.051 |
| 918  | Q0250   | COX2    | 0         | 0.002 | 0.045  | 0.051 |
| 919  | Q0275   | COX3    | 0         | 0.002 | 0.045  | 0.051 |
| 920  | YER010C | YER010C | 0         | 0.001 | -0.182 | 0.281 |
| 921  | YNL334C | SNO2    | 0         | 0.001 | -0.182 | 0.097 |
| 922  | YFL060C | SNO3    | 0         | 0.001 | -0.182 | 0.097 |
| 923  | YOL020W | TAT2    | 0         | 0.001 | -0.182 | 0.050 |
| 924  | YLR146C | SPE4    | 0         | 0.001 | -0.182 | 0.356 |
| 925  | YJL198W | PHO90   | 0         | 0.001 | -0.182 | 0.154 |
| 926  | YCR037C | PHO87   | 0         | 0.001 | -0.182 | 0.154 |
| 927  | YML123C | PHO84   | 0         | 0.001 | -0.182 | 0.154 |
| 928  | YBR204C | LDH1    | 0         | 0.001 | -0.182 | 0.078 |
| 929  | YCL069W | VBA3    | 0         | 0.001 | -0.182 | 0.126 |
| 930  | YMR088C | VBA1    | 0         | 0.001 | -0.182 | 0.126 |
| 931  | YBR281C | DUG2    | 0         | 0.001 | -0.182 | 0.209 |
| 932  | YNL191W | DUG3    | 0         | 0.001 | -0.182 | 0.209 |
| 933  | YDR046C | BAP3    | 0         | 0.001 | -0.182 | 0.043 |
| 934  | YBR068C | BAP2    | 0         | 0.001 | -0.182 | 0.043 |
| 935  | YAR071W | PHO11   | 0         | 0.001 | -0.182 | 0.177 |
| 936  | YBR093C | PHO5    | 0         | 0.001 | -0.182 | 0.177 |
| 937  | YHR215W | PHO12   | 0         | 0.001 | -0.182 | 0.177 |
| 938  | YGL084C | GUP1    | 0         | 0.001 | -0.182 | 0.214 |
| 939  | YPL189W | GUP2    | 0         | 0.001 | -0.182 | 0.214 |
| 940  | YHR051W | COX6    | 0         | 0.001 | -0.182 | 0.005 |
| 941  | YGL191W | COX13   | 0         | 0.001 | -0.182 | 0.005 |
| 942  | YGL187C | COX4    | 0         | 0.001 | -0.182 | 0.005 |
| 943  | YMR256C | COX7    | 0         | 0.001 | -0.182 | 0.005 |
| 944  | YLR395C | COX8    | 0         | 0.001 | -0.182 | 0.005 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name   | essential | GCN   | FBA    | LR    |
|------|-----------|--------|-----------|-------|--------|-------|
| 945  | YLR038C   | COX12  | 0         | 0.001 | -0.182 | 0.005 |
| 946  | YOL157C   | IMA2   | 0         | 0.001 | -0.182 | 0.371 |
| 947  | YHR037W   | PUT2   | 0         | 0.001 | -0.182 | 0.388 |
| 948  | YPL134C   | ODC1   | 0         | 0.001 | -0.182 | 0.204 |
| 949  | YOR222W   | ODC2   | 0         | 0.001 | -0.182 | 0.204 |
| 950  | YDR231C   | COX20  | 0         | 0.001 | -0.182 | 0.006 |
| 951  | YLL018C-A | COX19  | 0         | 0.001 | -0.182 | 0.006 |
| 952  | YPL132W   | COX11  | 0         | 0.001 | -0.182 | 0.006 |
| 953  | YJL003W   | COX16  | 0         | 0.001 | -0.182 | 0.006 |
| 954  | YHR116W   | COX23  | 0         | 0.001 | -0.182 | 0.006 |
| 955  | YGR062C   | COX18  | 0         | 0.001 | -0.182 | 0.006 |
| 956  | YHR002W   | LEU5   | 0         | 0.001 | -0.182 | 0.351 |
| 957  | YNL052W   | COX5A  | 0         | 0.001 | -0.182 | 0.005 |
| 958  | YIL111W   | COX5B  | 0         | 0.001 | -0.182 | 0.005 |
| 959  | YMR293C   | HER2   | 0         | 0.001 | -0.182 | 0.409 |
| 960  | YOR270C   | VPH1   | 0         | 0.001 | -0.182 | 0.163 |
| 961  | YMR054W   | STV1   | 0         | 0.001 | -0.182 | 0.163 |
| 962  | YCR048W   | ARE1   | 0         | 0.001 | -0.182 | 0.183 |
| 963  | YHR092C   | HXT4   | 0         | 0.000 | -0.182 | 0.224 |
| 964  | YLR158C   | ASP3-3 | 0         | 0.000 | -0.182 | 0.166 |
| 965  | YLR155C   | ASP3-1 | 0         | 0.000 | -0.182 | 0.166 |
| 966  | YLR160C   | ASP3-4 | 0         | 0.000 | -0.182 | 0.166 |
| 967  | YLR157C   | ASP3-2 | 0         | 0.000 | -0.182 | 0.166 |
| 968  | YFL059W   | SNZ3   | 0         | 0.000 | -0.182 | 0.150 |
| 969  | YNL333W   | SNZ2   | 0         | 0.000 | -0.182 | 0.150 |
| 970  | YER090W   | TRP2   | 0         | 0.000 | -0.182 | 0.172 |
| 971  | YER053C   | PIC2   | 0         | 0.000 | -0.182 | 0.226 |
| 972  | YJR077C   | MIR1   | 0         | 0.000 | -0.182 | 0.226 |
| 973  | YEL071W   | DLD3   | 0         | 0.000 | -0.182 | 0.080 |
| 974  | YDR105C   | TMS1   | 0         | 0.000 | -0.182 | 0.053 |
| 975  | YKL106W   | AAT1   | 0         | 0.000 | -0.182 | 0.301 |
| 976  | YPL212C   | PUS1   | 0         | 0.000 | -0.182 | 0.138 |
| 977  | YNL292W   | PUS4   | 0         | 0.000 | -0.182 | 0.138 |
| 978  | YFL001W   | DEG1   | 0         | 0.000 | -0.182 | 0.138 |
| 979  | YGL063W   | PUS2   | 0         | 0.000 | -0.182 | 0.138 |
| 980  | YER005W   | YND1   | 0         | 0.000 | -0.182 | 0.085 |
| 981  | YDL174C   | DLD1   | 0         | 0.000 | -0.182 | 0.047 |
| 982  | YFL055W   | AGP3   | 0         | 0.000 | -0.182 | 0.077 |
| 983  | YGL253W   | HXK2   | 0         | 0.000 | -0.182 | 0.095 |
| 984  | YFR053C   | HXK1   | 0         | 0.000 | -0.182 | 0.095 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene      | name  | essential | GCN   | FBA    | LR    |
|------|-----------|-------|-----------|-------|--------|-------|
| 985  | YGR247W   | CPD1  | 0         | 0.000 | -0.182 | 0.437 |
| 986  | YKL211C   | TRP3  | 0         | 0.000 | -0.182 | 0.167 |
| 987  | YHR208W   | BAT1  | 0         | 0.000 | -0.059 | 0.230 |
| 988  | YDL245C   | HXT15 | 0         | 0.000 | -0.182 | 0.147 |
| 989  | YBR299W   | MAL32 | 0         | 0.000 | -0.182 | 0.195 |
| 990  | YGR292W   | MAL12 | 0         | 0.000 | -0.182 | 0.195 |
| 991  | YDR508C   | GNP1  | 0         | 0.000 | -0.182 | 0.081 |
| 992  | YDR342C   | HXT7  | 0         | 0.000 | -0.182 | 0.138 |
| 993  | YDR533C   | HSP31 | 0         | 0.000 | -0.182 | 0.152 |
| 994  | YOR391C   | HSP33 | 0         | 0.000 | -0.182 | 0.152 |
| 995  | YDR516C   | EMI2  | 0         | 0.000 | -0.182 | 0.056 |
| 996  | YCL040W   | GLK1  | 0         | 0.000 | -0.182 | 0.056 |
| 997  | YHL036W   | MUP3  | 0         | 0.000 | -0.182 | 0.029 |
| 998  | YGR055W   | MUP1  | 0         | 0.000 | -0.182 | 0.029 |
| 999  | YOL068C   | HST1  | 0         | 0.000 | -0.182 | 0.110 |
| 1000 | YOR025W   | HST3  | 0         | 0.000 | -0.182 | 0.110 |
| 1001 | YPL015C   | HST2  | 0         | 0.000 | -0.182 | 0.110 |
| 1002 | YDR191W   | HST4  | 0         | 0.000 | -0.182 | 0.110 |
| 1003 | YDL042C   | SIR2  | 0         | 0.000 | -0.182 | 0.110 |
| 1004 | YEL051W   | VMA8  | 0         | 0.000 | -0.182 | 0.134 |
| 1005 | YBR127C   | VMA2  | 0         | 0.000 | -0.182 | 0.134 |
| 1006 | YGR106C   | VOA1  | 0         | 0.000 | -0.182 | 0.134 |
| 1007 | YEL027W   | VMA3  | 0         | 0.000 | -0.182 | 0.134 |
| 1008 | YHR039C-A | VMA10 | 0         | 0.000 | -0.182 | 0.134 |
| 1009 | YLR447C   | VMA6  | 0         | 0.000 | -0.182 | 0.134 |
| 1010 | YKL080W   | VMA5  | 0         | 0.000 | -0.182 | 0.134 |
| 1011 | YHR026W   | VMA16 | 0         | 0.000 | -0.182 | 0.134 |
| 1012 | YPL234C   | VMA11 | 0         | 0.000 | -0.182 | 0.134 |
| 1013 | YGR020C   | VMA7  | 0         | 0.000 | -0.182 | 0.134 |
| 1014 | YPR036W   | VMA13 | 0         | 0.000 | -0.182 | 0.134 |
| 1015 | YCL005W-A | VMA9  | 0         | 0.000 | -0.182 | 0.134 |
| 1016 | YDL185W   | VMA1  | 0         | 0.000 | -0.182 | 0.134 |
| 1017 | YOR332W   | VMA4  | 0         | 0.000 | -0.182 | 0.134 |
| 1018 | YER174C   | GRX4  | 0         | 0.000 | -0.182 | 0.125 |
| 1019 | YDR098C   | GRX3  | 0         | 0.000 | -0.182 | 0.125 |
| 1020 | YGR209C   | TRX2  | 0         | 0.000 | -0.182 | 0.046 |
| 1021 | YLR027C   | AAT2  | 0         | 0.000 | -0.125 | 0.133 |
| 1022 | YER037W   | PHM8  | 0         | 0.000 | -0.182 | 0.235 |
| 1023 | YBR298C   | MAL31 | 0         | 0.000 | -0.182 | 0.140 |
| 1024 | YER073W   | ALD5  | 0         | 0.000 | -0.182 | 0.113 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name  | essential | GCN   | FBA    | LR    |
|------|---------|-------|-----------|-------|--------|-------|
| 1025 | YDR536W | STL1  | 0         | 0.000 | -0.182 | 0.102 |
| 1026 | YCL035C | GRX1  | 0         | 0.000 | -0.182 | 0.106 |
| 1027 | YDR513W | GRX2  | 0         | 0.000 | -0.182 | 0.106 |
| 1028 | YJL060W | BNA3  | 0         | 0.000 | -0.182 | 0.104 |
| 1029 | YPL265W | DIP5  | 0         | 0.000 | -0.182 | 0.047 |
| 1030 | YDL247W | MPH2  | 0         | 0.000 | -0.182 | 0.073 |
| 1031 | YJR160C | MPH3  | 0         | 0.000 | -0.182 | 0.073 |
| 1032 | YCR107W | AAD3  | 0         | 0.000 | -0.182 | 0.057 |
| 1033 | YNL331C | AAD14 | 0         | 0.000 | -0.182 | 0.057 |
| 1034 | YDL243C | AAD4  | 0         | 0.000 | -0.182 | 0.057 |
| 1035 | YJR155W | AAD10 | 0         | 0.000 | -0.182 | 0.057 |
| 1036 | YOL165C | AAD15 | 0         | 0.000 | -0.182 | 0.057 |
| 1037 | YOR126C | IAH1  | 0         | 0.000 | -0.182 | 0.161 |
| 1038 | YNR013C | PHO91 | 0         | 0.000 | -0.182 | 0.220 |
| 1039 | YNL318C | HXT14 | 0         | 0.000 | -0.182 | 0.091 |
| 1040 | YEL069C | HXT13 | 0         | 0.000 | -0.182 | 0.058 |
| 1041 | YNR072W | HXT17 | 0         | 0.000 | -0.182 | 0.058 |
| 1042 | YJR158W | HXT16 | 0         | 0.000 | -0.182 | 0.058 |
| 1043 | YLR081W | GAL2  | 0         | 0.000 | -0.182 | 0.043 |
| 1044 | YMR169C | ALD3  | 0         | 0.000 | -0.182 | 0.162 |
| 1045 | YHR104W | GRE3  | 0         | 0.000 | -0.182 | 0.212 |
| 1046 | YDR368W | YPR1  | 0         | 0.000 | -0.182 | 0.109 |
| 1047 | YCR105W | ADH7  | 0         | 0.000 | -0.182 | 0.131 |
| 1048 | YMR318C | ADH6  | 0         | 0.000 | -0.182 | 0.131 |
| 1049 | YEL039C | CYC7  | 0         | 0.000 | -0.182 | 0.001 |
| 1050 | YJR048W | CYC1  | 0         | 0.000 | -0.182 | 0.001 |
| 1051 | YKR039W | GAP1  | 0         | 0.000 | 0.100  | 0.100 |
| 1052 | YCL025C | AGP1  | 0         | 0.000 | -0.182 | 0.023 |
| 1053 | YJL219W | HXT9  | 0         | 0.000 | -0.182 | 0.034 |
| 1054 | YOL156W | HXT11 | 0         | 0.000 | -0.182 | 0.034 |
| 1055 | YFL011W | HXT10 | 0         | 0.000 | -0.182 | 0.034 |
| 1056 | YGR282C | BGL2  | 0         | 0.000 | -0.182 | 0.073 |
| 1057 | YLR300W | EXG1  | 0         | 0.000 | -0.182 | 0.073 |
| 1058 | YNR067C | DSE4  | 0         | 0.000 | -0.182 | 0.073 |
| 1059 | YOR190W | SPR1  | 0         | 0.000 | -0.182 | 0.073 |
| 1060 | YDR261C | EXG2  | 0         | 0.000 | -0.182 | 0.073 |
| 1061 | YBR244W | GPX2  | 0         | 0.000 | -0.182 | 0.041 |
| 1062 | YKL026C | GPX1  | 0         | 0.000 | -0.182 | 0.041 |
| 1063 | YNL229C | URE2  | 0         | 0.000 | -0.182 | 0.041 |
| 1064 | YIR037W | HYR1  | 0         | 0.000 | -0.182 | 0.041 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name | essential | GCN   | FBA    | LR    |
|------|---------|------|-----------|-------|--------|-------|
| 1065 | YLR047C | FRE8 | 0         | 0.000 | -0.182 | 0.028 |
| 1066 | YOL152W | FRE7 | 0         | 0.000 | -0.182 | 0.028 |
| 1067 | YOR381W | FRE3 | 0         | 0.000 | -0.182 | 0.028 |
| 1068 | YNR060W | FRE4 | 0         | 0.000 | -0.182 | 0.028 |
| 1069 | YKL220C | FRE2 | 0         | 0.000 | -0.182 | 0.028 |
| 1070 | YLR214W | FRE1 | 0         | 0.000 | -0.182 | 0.028 |
| 1071 | YOR384W | FRE5 | 0         | 0.000 | -0.182 | 0.028 |
| 1072 | YDR483W | KRE2 | 0         | 0.000 | -0.182 | 0.020 |
| 1073 | YPL053C | KTR6 | 0         | 0.000 | -0.182 | 0.020 |
| 1074 | YNL029C | KTR5 | 0         | 0.000 | -0.182 | 0.020 |
| 1075 | YBR199W | KTR4 | 0         | 0.000 | -0.182 | 0.020 |
| 1076 | YJL139C | YUR1 | 0         | 0.000 | -0.182 | 0.020 |
| 1077 | YBR205W | KTR3 | 0         | 0.000 | -0.182 | 0.020 |
| 1078 | YKR061W | KTR2 | 0         | 0.000 | -0.182 | 0.020 |
| 1079 | YOR099W | KTR1 | 0         | 0.000 | -0.182 | 0.020 |
| 1080 | YPL061W | ALD6 | 0         | 0.000 | -0.182 | 0.208 |
| 1081 | YHR137W | ARO9 | 0         | 0.000 | -0.182 | 0.036 |
| 1082 | YLL052C | AQY2 | 0         | 0.000 | -0.182 | 0.043 |
| 1083 | YPR192W | AQY1 | 0         | 0.000 | -0.182 | 0.043 |
| 1084 | YGL202W | ARO8 | 0         | 0.000 | -0.182 | 0.037 |
| 1085 | YJL216C | IMA5 | 0         | 0.000 | -0.182 | 0.036 |
| 1086 | YIL172C | IMA3 | 0         | 0.000 | -0.182 | 0.036 |
| 1087 | YJL221C | IMA4 | 0         | 0.000 | -0.182 | 0.036 |
| 1088 | YGR287C | IMA1 | 0         | 0.000 | -0.182 | 0.036 |
| 1089 | YDL168W | SFA1 | 0         | 0.000 | -0.182 | 0.091 |
| 1090 | YMR170C | ALD2 | 0         | 0.000 | -0.182 | 0.078 |
| 1091 | YLR044C | PDC1 | 0         | 0.000 | -0.182 | 0.015 |
| 1092 | YGR087C | PDC6 | 0         | 0.000 | -0.182 | 0.015 |
| 1093 | YLR134W | PDC5 | 0         | 0.000 | -0.182 | 0.015 |
| 1094 | YFL018C | LPD1 | 0         | 0.000 | -0.176 | 0.031 |
| 1095 | YOR374W | ALD4 | 0         | 0.000 | -0.182 | 0.064 |
| 1096 | YGL256W | ADH4 | 0         | 0.000 | -0.182 | 0.064 |
| 1097 | YMR083W | ADH3 | 0         | 0.000 | -0.182 | 0.064 |
| 1098 | YBR145W | ADH5 | 0         | 0.000 | -0.182 | 0.045 |
| 1099 | YOL086C | ADH1 | 0         | 0.000 | -0.182 | 0.045 |
| 1100 | YDR242W | AMD2 | 0         | 0.000 | -0.182 | 0.130 |
| 1101 | YHR094C | HXT1 | 0         | 0.000 | -0.182 | 0.000 |
| 1102 | YDR345C | HXT3 | 0         | 0.000 | -0.182 | 0.000 |
| 1103 | YHR096C | HXT5 | 0         | 0.000 | -0.182 | 0.000 |
| 1104 | YJL214W | HXT8 | 0         | 0.000 | -0.182 | 0.000 |

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

| rank | gene    | name | essential | GCN   | FBA    | LR    |
|------|---------|------|-----------|-------|--------|-------|
| 1105 | YDR343C | HXT6 | 0         | 0.000 | -0.182 | 0.000 |
| 1106 | YMR011W | HXT2 | 0         | 0.000 | -0.182 | 0.000 |
| 1107 | YMR110C | HFD1 | 0         | 0.000 | -0.182 | 0.008 |

## F External tools and dataset manifest

Honest counts: 17 external tools and 10 accession-level datasets (`results/external_tools_manifest.csv`, `results/dataset_manifest.csv`). Dataset counting follows the program rule: one study's condition matrix is one dataset (Biolog = 1 dataset with 383 conditions; each chemostat compendium = 1). The manifests list every tool and dataset with version/accession, condition counts, and where each is used.